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Human-in-the-Loop Agenten in der Pflegerobotik: Evaluierung und interaktive Anwendung von OCR und VLMs zur Verarbeitung handschriftlicher Pflegedokumentation

Human-in-the-Loop Agents in Nursing Robotics: Evaluation and Interactive Application of OCR and VLMs for Processing Handwritten Care Records

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Kurzfassung

Die Verarbeitung handschriftlicher Pflegedokumentation stellt im klinischen Alltag eine erhebliche Herausforderung dar. Die manuelle Übertragung dieser Informationen in digitale Systeme ist nicht nur zeitaufwendig und ressourcenintensiv, sondern auch äußerst anfällig für Übertragungsfehler. Während traditionelle Methoden der optischen Zeichenerkennung (OCR) gedruckten Text effektiv digitalisieren, weisen sie bei der Anwendung auf handschriftliche, semantisch dichte und medizinisch spezialisierte Dokumente gravierende Einschränkungen auf. Um diesen Engpass zu überwinden, gewinnen multimodale Ansätze – insbesondere Vision-Language Models (VLMs) – aufgrund ihrer Fähigkeit, visuelle Inhalte und linguistischen Kontext gemeinsam zu modellieren, zunehmend an Bedeutung.

Diese Studie untersucht und vergleicht systematisch die Eignung entkoppelter OCR-LLM-Pipelines gegenüber hochmodernen End-to-End-VLMs zur Automatisierung der klinischen Informationsextraktion. Evaluert an einem stratifizierten Datensatz komplexer Pflegedokumente, zeigten die End-to-End-VLMs eine deutliche architektonische Überlegenheit, indem sie die sich aufsummierenden Kaskadenfehler eliminierten, die zweistufigen OCR-Pipelines innewohnen. Bemerkenswert ist, dass Qwen3-VL-8B und Pixtral-8B die besten Pipeline-Konfigurationen übertrafen und F1-Scores für medizinische Begriffe von 64,14 % bzw. 62,47 % erreichten.

Die Evaluierung deckte jedoch eine kritische Schwachstelle in aktuellen Zero-Shot VLM-Anwendungen auf: eine extreme Leistungsvarianz über verschiedene Dokumentenlayouts hinweg, wobei individuelle Extraktionswerte zwischen 0,0 % und 100,0 % schwankten. Diese hohe Varianz unterstreicht die anhaltende Notwendigkeit eines „Human-in-the-Loop“-Verifikationsschritts, um Unsicherheiten zu reduzieren und die Zuverlässigkeit in sensiblen klinischen Kontexten zu gewährleisten.

Letztendlich erforderte die Überführung dieser theoretischen Metriken in einen praktischen Proof-of-Concept für eine mobile Roboterplattform ein Abwägen zwischen maximaler algorithmischer Leistung und strengen hardwaretechnischen sowie rechtlichen Einschränkungen. Um die Einhaltung der Datenschutz-Grundverordnung (DSGVO) zu gewährleisten, ohne die lokalen Rechenkapazitäten des Roboters (Onboard-Ressourcen) zu überschreiten, wurde das Modell Pixtral-8B – angebunden über eine in Europa gehostete Enterprise-API – ausgewählt. Diese Arbeit zeigt, dass End-to-End-VLMs zwar einen bedeutenden Fortschritt im maschinellen Dokumentenverständnis für die Robotik darstellen, der praktische Einsatz jedoch voraussetzt, dass Datensouveränität und Systemkonsistenz ebenso entscheidend sind wie die reine Extraktionsgenauigkeit.

Abstract

The processing of handwritten nursing documentation poses a significant challenge in everyday clinical practice. The manual transfer of this information into digital systems is not only time-consuming and resource-intensive, but also highly susceptible to transcription errors. While traditional Optical Character Recognition (OCR) methods effectively digitize printed text, they demonstrate severe limitations when applied to handwritten, semantically dense, and medically specialized documents. To address this bottleneck, multimodal approaches—specifically Vision-Language Models (VLMs)—are gaining prominence due to their ability to jointly model visual content and linguistic context.

This study systematically investigates and compares the suitability of decoupled OCR-LLM pipelines against state-of-the-art, end-to-end VLMs for automating clinical information extraction. Evaluated on a stratified dataset of complex nursing documents, the end-to-end VLMs demonstrated clear architectural superiority by eliminating the compounding cascading errors inherent to two-step OCR pipelines. Notably, Qwen3-VL-8B and Pixtral-8B surpassed the best pipeline configurations, achieving Medical Term F1-scores of 64.14% and 62.47%, respectively.

However, the evaluation exposed a critical vulnerability in current zero-shot VLM deployments: extreme performance variance across different document layouts, with individual extraction scores oscillating between 0.0% and 100.0%. This high variance highlights the ongoing necessity of a "Human-in-the-Loop" verification step to mitigate uncertainties and ensure reliability in sensitive clinical contexts.

Ultimately, translating these theoretical metrics into a practical proof-of-concept for a mobile robotic platform required balancing peak algorithmic performance with stringent hardware and legal constraints. To ensure compliance with the General Data Protection Regulation (GDPR) without exceeding onboard computational limits, the Pixtral-8B model—accessed via a European-hosted enterprise API—was selected. This work demonstrates that while end-to-end VLMs represent a significant leap forward in robotic document understanding, practical deployment dictates that data sovereignty and system consistency remain equally critical to raw extraction accuracy.

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1. Introduction

1.1. Motivation

The processing of handwritten nursing documentation continues to pose a significant challenge in everyday clinical and nursing practice. In many operational contexts, documentation is still performed manually, which complicates subsequent digital processing. The manual transfer of this information into digital systems is not only time-consuming but also consumes personnel resources that are already scarce in daily nursing operations. Simultaneously, the risk of transcription errors, omissions, and misinterpretations increases—particularly when entries are incomplete, highly condensed, or difficult to decipher due to individual handwriting styles.

To date, OCR-based methods have frequently been employed for the automated processing of such documents. While these methods often yield good results for printed texts, they demonstrate distinct limitations when applied to handwritten, semantically dense, and medically specialized documents. In particular, the combination of variable handwriting, domain-specific terminology, and context-dependent information places high demands on machine recognition and interpretation. Against this backdrop, multimodal approaches are gaining prominence, as Vision-Language Models (VLMs) are capable of not merely extracting characters and words, but of jointly modeling both visual content and linguistic context.

The primary objective of this study is to systematically investigate the suitability of various processing approaches for handwritten nursing documentation. To this end, decoupled OCR-based pipelines and end-to-end VLM methods are compared against one another. To systematically guide this evaluation, the following central research questions are addressed:

1. **Extraction Accuracy:**

To what extent do VLM models improve the extraction accuracy of handwritten nursing documentation compared to a purely OCR-based approach?

2. **Semantic Comprehension and Summarization:**

Do VLMs offer superior contextual understanding and generate higher-quality clinical summaries than the traditional combination of classical LLMs with decoupled OCR engines?

3. **Reliability in Critical Domains:**

How effectively do the different architectural approaches mitigate hallucinations when confronted with complex handwriting, lengthy texts, and highly specific medical terminology?

Furthermore, recognizing the inherent risks of fully automated data entry in healthcare, this study analyzes the extent to which a “Human-in-the-Loop” agentic workflow can help to specifically mitigate uncertainties, facilitate interactive queries, and thereby enhance the overall system’s reliability and practical utility within sensitive operational contexts.

1.2. Related Work

The problem of extracting accurate, structured data from complex medical and clinical documents has been approached from multiple angles in recent literature. While traditional Optical Character Recognition (OCR) systems have served as the baseline for digitization, modern research has increasingly focused on mitigating OCR errors through Large Language Models (LLMs) or bypassing traditional OCR entirely using Vision-Language Models (VLMs). This chapter reviews recent advancements in both paradigms and highlights the current gaps in nursing and clinical data extraction.

1.2.1. LLMs for OCR Post-Correction

The inherent noise in clinical documents—caused by poor scanning quality, complex tabular layouts, and specialized medical abbreviations—frequently leads to high Character Error Rates (CER) in standard OCR outputs. Consequently, a prominent area of research focuses on using LLMs as a secondary post-processing step to correct and structure OCR output.

Veninga (1) investigated the efficacy of character-level LLMs, such as ByT5, to directly fix OCR mistakes. The study demonstrated that while standard generative LLMs struggle with zero-shot OCR correction due to hallucination risks, carefully designed context windows and targeted prompting significantly reduce error rates. Building upon this, Kang et al. (2) developed a specialized Natural Language Processing (NLP) pipeline explicitly designed for clinical note correction. Their work proved that applying LLMs to correct highly specialized medical OCR outputs (such as karyotype data) provides substantial improvements over baseline deep learning OCR models.

Furthermore, recent studies have evaluated complete sequential pipelines. For instance, research on pathology report digitization (3) evaluated a multi-engine architecture where an OCR system extracts raw text, which is subsequently passed to an LLM for error correction and JSON structuring. These studies establish the OCR-to-LLM pipeline as a highly viable, state-of-the-art approach for handling unstructured medical data.

1.2.2. VLMs for Direct Clinical Document Understanding

While OCR combined with LLM pipelines improves text accuracy, it suffers from a fundamental “segmentation paradox”: the LLM cannot recover spatial or layout context that the OCR engine failed to capture. To address this, recent literature has shifted toward Vision-Language Models (VLMs) that analyze the document image and text concurrently.

A multimodal pipeline introduced recently (4) applied VLMs to extract data directly from clinical forms, specifically targeting complex elements like checkboxes and unstructured transfusion reaction reports. Their findings indicated that combining bounding box intelligence with VLM reasoning is highly effective for complex clinical layouts. More recently, the MeDocVL framework (5) addressed the specific limitations of general-purpose VLMs. The authors noted that standard OCR and broad-domain VLMs frequently fail on medical documents due to dense layouts and niche terminology, prompting the development of a domain-specific visual language model that outperformed conventional parsing systems.

Finally, recent benchmarking studies (6) comparing traditional OCR engines against modern VLMs on noisy scanned forms found that VLMs achieved significantly higher structural accuracy

on complex layouts. However, this accuracy came at a steep computational cost, with VLMs operating substantially slower than their OCR counterparts.

1.2.3. Contribution of this Thesis

Current literature clearly establishes two distinct pathways for clinical document digitization: the sequential OCR-to-LLM correction pipeline and the end-to-end VLM approach. While existing studies have benchmarked these models on general scanned forms or highly specific pathology reports, there remains a gap regarding their direct comparison on complex, multimodal nursing records.

This thesis bridges that gap by directly benchmarking a disjointed architecture (EasyOCR, Surya and LightOnOCR paired with an LLM) against a unified Vision-Language Model (Qwen3VL, Pixtral and Chatgpt4o-mini). By evaluating these systems specifically on clinical and nursing documents, this work contributes a critical analysis of the current accuracy-versus-speed trade-offs in modern medical document processing.

1.3. Key Concepts and Terminology

This section defines the key terms and concepts used in this paper to ensure a consistent understanding of the topic.

1.3.1. Optical Character Recognition (OCR)

Optical character recognition (OCR) is an information technology process that analyzes image data—such as scanned physical documents, photographs, or PDF files—and converts the text information contained therein into machine-readable, editable, and searchable character strings. For a computer, a digitized document initially consists exclusively of a matrix of pixels that do not carry any semantic text information. The task of an OCR system is to extract structured text data from this unstructured visual data.(7)

1.3.1.1. Traditional OCR: The Heuristic Approach

early OCR systems were based on strictly sequential, rule-based architectures. In these heuristic systems, the process from static image to digital text takes place in a deterministic pipeline consisting of several consecutive processing steps (8):

- **Image capture and binarization:** In the first step, the scanned document is converted into a pure black-and-white representation. Using methods such as adaptive thresholding, the algorithm separates the dark areas as text (foreground) from the light areas as background in order to classify dark areas as text and light areas as background (9).
- **Image pre-processing:** To optimize the subsequent recognition rate, the binarized image is cleaned up algorithmically. Typical operations include deskewing, contrast enhancement, and despeckling (9).
- **Layout analysis and segmentation:** The system analyzes the visual structure of the document and isolates the text from graphic elements. Segmentation then takes place according to the “top-down principle”: The text block is divided into lines, the lines into words, and the words into individually isolated characters. This separation is usually done by analyzing empty spaces (connected component analysis) (10).

- **Text recognition (classification):** In this core step, the isolated character is identified. In traditional OCR, this is primarily done through pattern matching or feature extraction (8). In pattern matching, the character is compared pixel by pixel with an internal database of known standard fonts (templates). In feature extraction, the system checks the character for hard-coded geometric properties, such as the presence of lines, intersections, or closed loops.
- **Post-processing:** Finally, the generated text is validated. Static lexicons or statistical models (such as N-grams) compare the recognized words with existing vocabulary to correct typical recognition errors—such as confusing the number “0” with the letter “O”—based on linguistic rules (11).

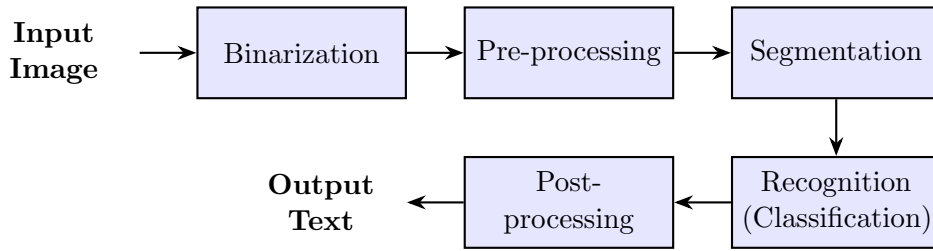


Figure 1.1.: The strictly sequential pipeline of traditional heuristic OCR systems.

Limitations: The Boundaries of Traditional OCR in Handwriting While the sequential pipeline described above achieves high recognition rates for machine-printed text (standard OCR), it reaches its conceptual limits when recognizing handwriting—referred to in the literature as *Handwritten Text Recognition* (HTR). The error susceptibility of traditional systems in HTR can primarily be attributed to two factors:

1. **The Segmentation Paradox:** The traditional pipeline strictly requires the isolation of individual characters prior to actual recognition. However, handwriting, particularly connected cursive script, cannot be reliably segmented into discrete characters due to fluid transitions and overlapping strokes (ligatures). An error made in this early segmentation phase inevitably propagates through to the recognition phase, compounding the final error rate. This fundamental dilemma—that a cursively written word cannot be segmented without being recognized, and cannot be recognized without being segmented—is known in the literature as *Sayre’s Paradox* (12; 13).
2. **Intra- and Inter-Writer Variability:** Unlike printed text, there is no standardized typography for handwriting that could serve as a template for pattern matching. Individual writing styles vary significantly not only between different individuals (*inter-writer variability*) but also within the handwriting of the same person across different contexts and times (*intra-writer variability*) (13; 14). These limitations clearly indicate that rule-based heuristic approaches are insufficient for handwritten text recognition. This realization led recent research to a paradigm shift towards holistic, deep learning-based end-to-end architectures that overcome the segmentation problem by processing words or even entire lines as undivided sequences (15; 16).

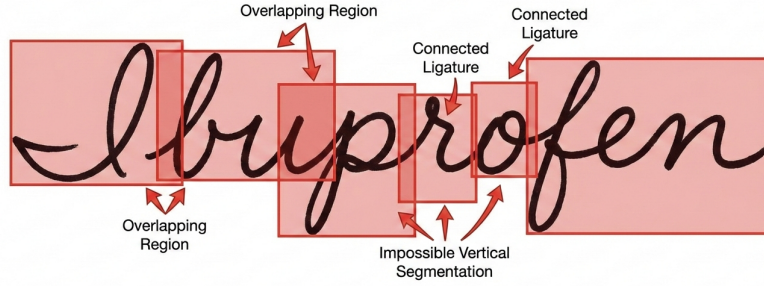


Figure 1.2.: The Segmentation Paradox in Cursive OCR: To recognize the letters, the system must first segment the word; however, to correctly segment the word, the system must already recognize the letters. As shown in the example of the medical term “Ibuprofen”, fluid ligatures make it impossible to draw discrete, non-overlapping bounding boxes for vertical segmentation (17).

1.3.1.2. Modern OCR: Deep Learning Approach

In contrast to traditional, rule-based OCR pipelines, modern OCR systems replace heuristic rules and template-based matching with trainable deep learning models that optimize the complete recognition objective from data. Instead of requiring perfectly binarized inputs and explicit character-wise segmentation, neural architectures learn invariances to common degradations (e.g., blur, skew, cluttered backgrounds) and model text as a sequence, which is particularly relevant for handwritten text recognition. Modern deep learning approaches generally follow a two-stage pipeline:

- **Text Detection with CRAFT:** The *Character Region Awareness for Text Detection* (CRAFT) module is a convolutional neural network (CNN) that does not search for rigid lines. Instead, it estimates the probability of whether a pixel belongs to a character or the space between characters. This allows it to robustly enclose even curved, slanted, or distorted text (e.g., on road signs or in photos) in bounding boxes (18).
- **Text Recognition with CRNN:** The contents of these boxes are then passed to a Convolutional Recurrent Neural Network (CRNN). This architecture usually consists of a CNN backbone (such as ResNet) for feature extraction, followed by bidirectional Long Short-Term Memory (LSTM) layers. The key feature here is that the network does not merely recognize isolated letters but understands the text as a sequence. By utilizing a Connectionist Temporal Classification (CTC) decoder, the system captures the word as a whole and avoids errors caused by closely spaced or overlapping letters (19).

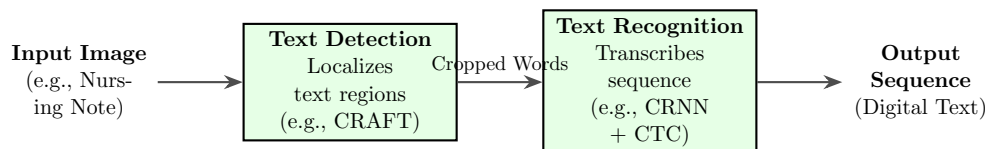


Figure 1.3.: Architecture of a modern two-stage Deep Learning OCR system, processing text at the sequence level without explicit character segmentation. Based on the concepts of CRAFT (18) and CRNN (19).

1.3.1.3. Next-Generation Architectures: Transformers and VLMs

Beyond this established two-stage pipeline, recent research has introduced two paradigm-shifting architectures that increasingly replace traditional OCR methods:

- **Transformer-based OCR (e.g., Surya OCR, Donut):** Moving away from complex CNN/RNN combinations, these models utilize a pure encoder-decoder Transformer architecture. They treat document image understanding and text extraction as a unified process, benefiting from much stronger language-context modeling and layout parsing capabilities compared to standard CRNN pipelines (20; 21).
- **Vision Language Models (VLMs):** The most radical departure from traditional pipelines is the use of multimodal foundation models (e.g., Qwen-VL or GPT-4V). Instead of relying on a dedicated OCR module to extract text, VLMs process the raw image directly and generate semantically structured text in a single step (OCR-free approach). Because they perform recognition and semantic reasoning simultaneously, VLMs are exceptionally robust to noise, complex layouts, and highly variable handwriting, overcoming the limits of both traditional and early deep learning OCR systems (22).

1.3.2. Large Language Models (LLMs)

While Optical Character Recognition (OCR) systems excel at converting images into raw text, they lack true cognitive understanding of the extracted data (7). To bridge the gap between unstructured character extraction and structured information processing, modern document pipelines integrate Large Language Models (LLMs) as downstream reasoning engines.

1.3.2.1. Tokenization and Probabilistic Generation

At their core, LLMs do not "read" words; they process text mathematically. A tokenizer first splits raw text into smaller units called *tokens* (sub-words or characters). The model then calculates a massive probability distribution over its entire vocabulary to determine the mathematically most likely next token (23). By repeatedly selecting the highest-probability token, the model generates fluent, human-like text.

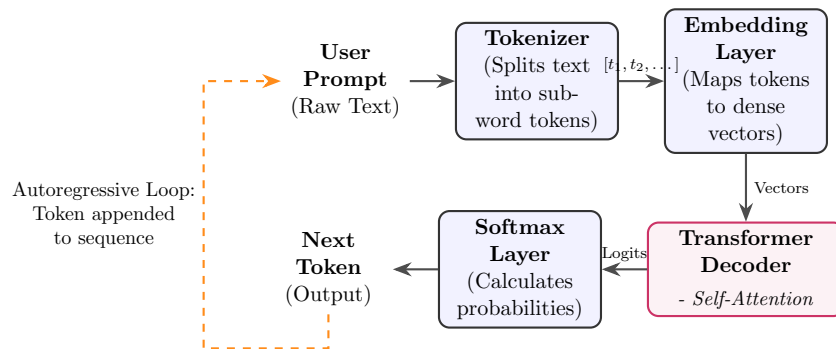


Figure 1.4.: Fundamental architecture and inference process of a Large Language Model (LLM). Raw text is tokenized and embedded into high-dimensional vectors before being processed by Transformer decoder blocks.

1.3.2.2. Transformer Decoders and Semantic Understanding

Modern LLMs are built upon the Transformer architecture, utilizing self-attention mechanisms to weigh the importance of all words in a sequence (24). Unlike traditional Natural Language Processing (NLP) pipelines that historically relied on rigid, hand-crafted grammar rules, syntax parsing, and predefined templates (25), LLMs develop a deep, zero-shot semantic understanding of language during their massive pre-training phase. This allows them to infer context, recognize complex relationships between data points, and understand the overarching narrative of a document without requiring domain-specific fine-tuning (23).

1.3.2.3. Context Windows and Information Integration

A critical capability of an LLM in document processing is its *context window*—the maximum amount of text the model can hold in memory and process simultaneously. Recent advancements have expanded context windows significantly, allowing LLMs to process entire multi-page OCR transcripts at once (26). By ingesting the full OCR output as a single prompt, the LLM can cross-reference data points and maintain structural coherence across long documents. For example, when processing a lengthy nursing admission record, the LLM can accurately connect a patient’s reported penicillin allergy on page one with a medication administration order on page five, synthesizing a cohesive clinical summary that a traditional fragmented OCR pipeline would miss (27).

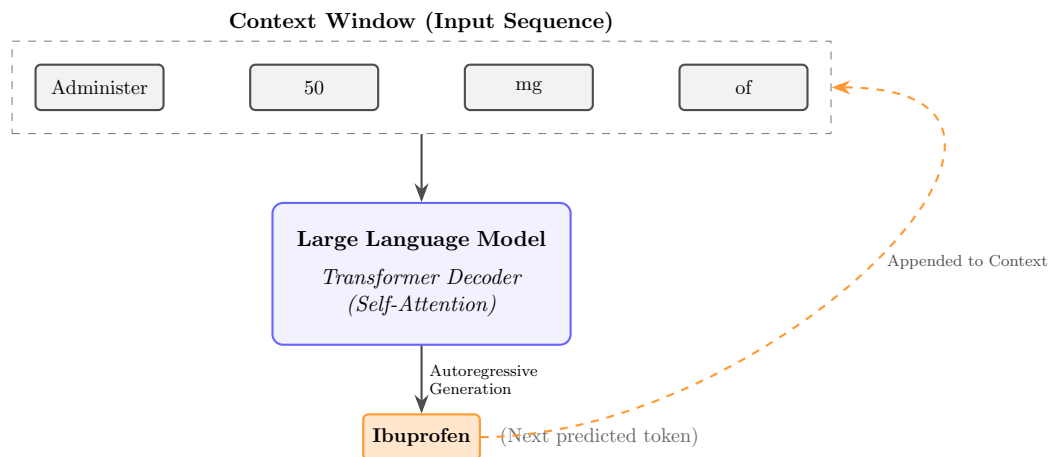


Figure 1.5.: Theoretical mechanism of autoregressive generation in an LLM applied to clinical text. The model utilizes self-attention to predict the highest-probability next token (“Ibuprofen”). The new token is then appended to the sequence to predict the subsequent word.

1.3.2.4. Autoregressive Reasoning and Summarization

Once the OCR text is loaded into the context window, the LLM generates its response through autoregressive decoding, predicting the most logical next token based on the provided text (23). Because the model inherently understands language structure, it can be prompted to perform highly complex cognitive tasks on the raw OCR data. Instead of merely transcribing text, the LLM actively reasons over the input to summarize lengthy records, extract specific key-value pairs into structured machine-readable formats (e.g., JSON), or classify document types based on semantic content (28).

1.3.2.5. Contextual Error Correction (Post-OCR Inference)

Raw OCR outputs frequently contain spelling mistakes, fragmented words, and misrecognized characters (e.g., confusing "0" with "O" or "1" with "l") due to noisy image inputs or complex handwriting. While traditional post-processing relies on static dictionaries, LLMs leverage their vast linguistic pre-training to perform dynamic, context-aware error correction (29). For instance, if an OCR system extracts the garbled phrase "AdnInister 50 ng of Ihupr0fen," an LLM recognizes the semantic context of a medical or nursing note and autoregressively corrects it to "Administer 50 mg of Ibuprofen." This ability to infer the correct word based on the surrounding sentence structure significantly improves the final accuracy of the document pipeline without requiring changes to the underlying OCR engine (30).

Limitations: Hallucinations in Clinical Contexts

Despite their advanced reasoning capabilities, LLMs are fundamentally probabilistic models. A well-documented limitation is their tendency to "hallucinate"—generating text that is grammatically correct and highly plausible, but factually false or unsupported by the input data (31). In high-stakes domains such as healthcare and nursing, hallucinations pose a severe risk, as an LLM might invent a clinical symptom or alter a medication dosage while attempting to correct an OCR error. Consequently, integrating LLMs into clinical document pipelines requires careful *prompt engineering* to ground the model's outputs strictly within the provided OCR context, ensuring that the system acts as a faithful summarizer rather than a creative generator (32).

1.3.3. Vision Language Models (VLMs)

Historically, document processing required disjointed pipelines: an OCR engine to extract text, followed by an NLP model to interpret it. Vision-Language Models (VLMs) represent a paradigm shift by merging visual perception and semantic reasoning into a single, unified architecture (33). By processing raw pixels and natural language simultaneously, VLMs effectively eliminate the need for traditional OCR bounding-box detection, enabling end-to-end document understanding.

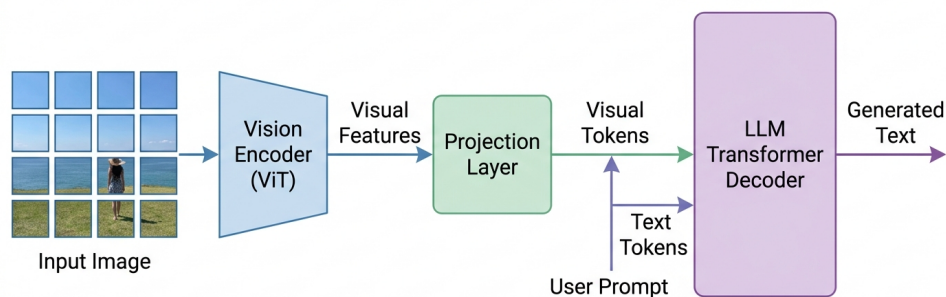


Figure 1.6.: The foundational architecture of a Vision-Language Model (VLM). An input image is divided into patches and processed by a Vision Encoder. The resulting visual features are passed through a projection layer, converting them into visual tokens that share the same embedding space as text tokens. These merged modalities are then processed autoregressively by the LLM backbone (34).

1.3.3.1. Architectural Fusion: Bridging Vision and Language

The foundational challenge of a VLM is translating unstructured visual data (pixels) into the discrete mathematical space (tokens) understood by a language model. This is typically achieved using a two-part architecture. First, a *Vision Encoder*—often based on a Vision Transformer (ViT)—divides the input image into a grid of patches and extracts high-dimensional visual feature vectors from each patch (35). Second, an *Embedding Projection Module* (such as a multi-layer perceptron or cross-attention layer) maps these visual features directly into the text embedding space of the LLM (33).

1.3.3.2. Multimodal Autoregressive Generation

Once the image patches are projected into the LLM’s embedding space, the model treats them as “visual words.” The LLM’s Transformer decoder can then process a combined sequence of text tokens (e.g., a user prompt like “Extract the patient’s name”) and visual tokens (the document image) (22). Utilizing the same autoregressive generation discussed in Section 1.3.2, the model predicts the most logical text output conditioned on both the text instructions and the visual context, effectively performing OCR and data extraction in a single step.

1.3.3.3. Spatial Reasoning and Layout Awareness

Unlike traditional OCR models that strip away formatting, or standard LLMs that only process flat text, VLMs inherently preserve spatial relationships. Because the vision encoder processes the document as a 2D grid, the VLM understands the physical layout of the page. This grants the model advanced spatial reasoning capabilities, allowing it to accurately parse complex, non-linear document structures such as multi-column clinical notes, tables, and handwritten annotations inserted between typed lines (20).

1.3.3.4. Robustness to Visual Degradation

Traditional OCR systems are highly sensitive to image noise, relying heavily on pre-processing techniques like binarization and deskewing to function accurately. In contrast, VLMs benefit from the massive scale of their multimodal pre-training. Because the vision encoder learns global image context rather than hard-coded character shapes, VLMs demonstrate exceptional zero-shot robustness against common document degradations such as low resolution, heavy scanning artifacts, varying lighting conditions, and highly irregular cursive handwriting (36).

Limitations: Sensitivity to Image Quality

While Vision-Language Models represent the current state-of-the-art in document understanding, this unified architecture introduces significant operational trade-offs. The most prominent limitation is their immense resource requirement; because the LLM backbone processes dense visual tokens alongside text, VLMs require substantial GPU VRAM for inference, making them significantly slower and more expensive to run than lightweight, specialized OCR pipelines like EasyOCR (37). Furthermore, while VLMs excel at understanding global layouts, their Vision Encoders are highly sensitive to low-level image degradation. If a scanned document is heavily pixelated, out of focus, or suffers from poor lighting and low contrast, the encoder struggles to extract meaningful features from the image patches. In such cases, the model may fail to resolve fine-grained details—such as distinguishing between a comma and a period, or an ‘l’ and a ‘1’—and is prone to hallucinating plausible but incorrect text based on semantic guesses rather than actual visual evidence (38).

2. Materials and Methods

2.1. Experimental Setup

To ensure the reproducibility of the proposed robotic pipeline, the physical experiments and model evaluations were conducted on a standardized hardware and software environment.

Hardware Infrastructure:

The core inference engine—responsible for both the local *Vision Routing Agent* and the subsequent on-device extraction baselines—was deployed on a desktop workstation. The system is equipped with an **Intel Core i5-13400F** processor, **32 GB of DDR5 RAM**, and an **NVIDIA GeForce RTX 3060 GPU with 12 GB of VRAM**. The available VRAM represents a strict constraint for local edge inference, deliberately chosen to simulate the computational limits typical of high-end mobile robotic platforms.

Software Environment and Virtualization:

Instead of a native Linux installation, the entire machine learning stack was isolated and executed within the **Windows Subsystem for Linux (WSL2)**. This approach allows for seamless hardware acceleration (CUDA passthrough) while maintaining a controlled, sandboxed environment for dependency management. A dedicated Python virtual environment was created to manage the extensive list of packages required for vision-language modeling, optical character recognition, and evaluation metrics.

2.2. Datasets

In order to meet the strict requirements for data protection and privacy in German geriatric care and to address the lack of freely available training data, a comprehensive synthetic dataset was generated. The creation process was divided into two phases: the technical validation of fonts and the construction of a clinically realistic text corpus.

2.2.1. Font Selection and Validation

A common challenge when utilizing freely available handwritten fonts is the inadequate support for the German character set. Specifically, German umlauts and special characters are frequently omitted from English-centric font packages.

To ensure the visual realism and orthographic correctness of the synthetic dataset, a strict quality assurance protocol was applied to all potential fonts.

The primary requirement was full encoding support for the character set prevalent in German medical documentation. This specifically encompasses:

- German umlauts (ä, ö, ü, Ä, Ö, Ü) and the Eszett (ß),
- Numeric digits (0–9),
- Common punctuation and medical special characters (e.g., -, !, ?, /, &, %).

To systematically verify this, a programmatic analysis was conducted using the Python library `fontTools`. A custom script was developed to extract and evaluate the Character Map

(Cmap) of 37 candidate fonts residing in the project’s directory, checking them strictly against the defined character set.

The automated analysis revealed significant discrepancies in character support across the candidates. Out of the 37 evaluated fonts, 13 were unable to render one or more essential characters (most frequently missing specific umlauts).

These were consequently excluded from the generation pipeline to prevent encoding errors or visual artifacts.

The remaining 24 fonts demonstrated full compatibility for all tested characters. These 24 validated fonts form the visual foundation for the synthetic document generation, ensuring that the resulting images fully comply with German handwriting standards.

2.2.2. Textual Corpus and Scenario Design

The textual component of the dataset consists of 50 unique care scenarios designed to simulate realistic documentation in inpatient geriatric care.

Metadata and Structure

All scenarios are structured as JSON objects containing specific metadata to enable automated processing and label creation. The metadata inherently reflects geriatric demographics, featuring realistic names and ages ranging from 71 to 94 years. Contextual data such as room numbers (01–50) and dates (October–November 2025) are assigned to each case. Furthermore, the dataset ensures a stratified distribution across the German care levels (Pflegegrade) 1 to 5 to accurately represent various patient conditions.

Clinical Content and Realism

The text corpus was specifically authored to reflect the linguistic characteristics of professional nursing reports. This includes a mixture of formal clinical terminology and the keyword-based, often condensed style commonly used in daily documentation. The texts incorporate common medical abbreviations (e.g., RR for blood pressure, BZ for blood sugar, SpO2 for oxygen saturation) and cover a wide range of clinical situations:

- **Acute & complex events:** Cognitive impairment and dementia (8 cases), infections (5 cases), falls (3 cases), and palliative care (3 cases).
- **Medical interventions:** Documentation of vital signs, medication administration, wound care (e.g., pressure ulcer classification), and pain management.
- **Routine care:** Daily nursing care, resident visits, and therapeutic measures (15 cases).

Data Stratification

To ensure high variance and realism in terms of text length and content complexity, the data set was strictly stratified according to care levels (CL). In the created corpus, this classification correlates directly with the required scope of documentation:

- **CL 1 (Low complexity):** Approx. 200–350 characters. These scenarios simulate largely independent residents. Documentation is limited to short, routine notes (e.g., “All meals taken in the dining room”).

2. Materials and Methods

- **CL 2–3 (Medium complexity):** Approx. 400–800 characters. This represents standard care documentation, including medium-length descriptions of daily activities, basic care, and medication administration.
- **CL 4–5 (High complexity):** Approx. 1,500–2,500+ characters. These scenarios include detailed medical reports for patients requiring intensive care. The content covers vital signs, positioning plans, wound documentation (e.g., pressure ulcers), PEG tube feeding, or palliative care.

By combining the 50 developed text scenarios with the 24 validated handwritten fonts, the system is capable of generating a highly diverse dataset comprising approximately 1,100 unique synthetic document images.

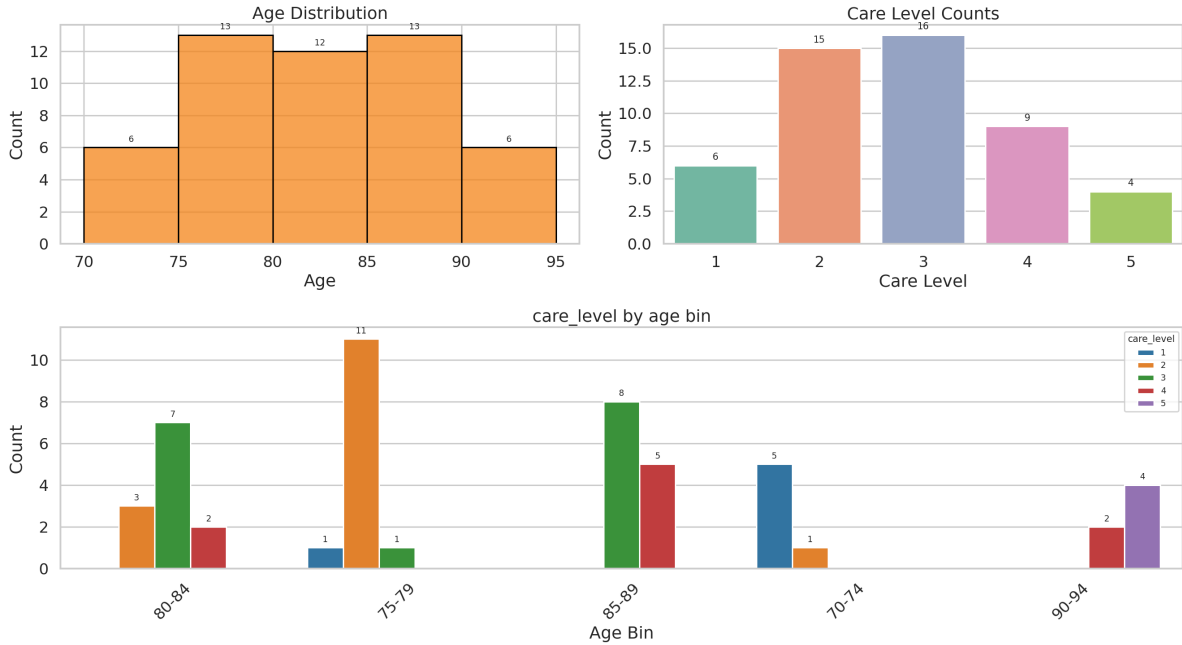


Figure 2.1.: Distribution of generated scenarios across patient age groups and care levels, demonstrating a realistic demographic spread for geriatric inpatient care.

2.2.3. Implementation of Data Generation

The generation pipeline was developed as a modular Python framework utilizing the Pillow (PIL) library for image processing. This fully automated process transforms the textual scenarios and validated fonts into annotated, realistic image data.

Visual Synthesis and Paper Simulation

To achieve a high degree of visual variation, the system dynamically generates the image background rather than relying on static templates. A heuristic algorithm (`select_page_profile_like_human`) analyzes the character length of the input text and autonomously selects an appropriate paper format. For instance, DIN A5 is chosen for short notes under 500 characters, while DIN A4 is utilized for detailed, complex reports. This directly imitates the natural format choices of nursing staff.

For each generation step, a weighted random selection assigns different paper colors (e.g., pure white, recycled gray, waste paper yellow) and ruling types (blank, lined, squared, dotted). The ruling types are drawn mathematically precisely on the canvas to avoid scaling artifacts.

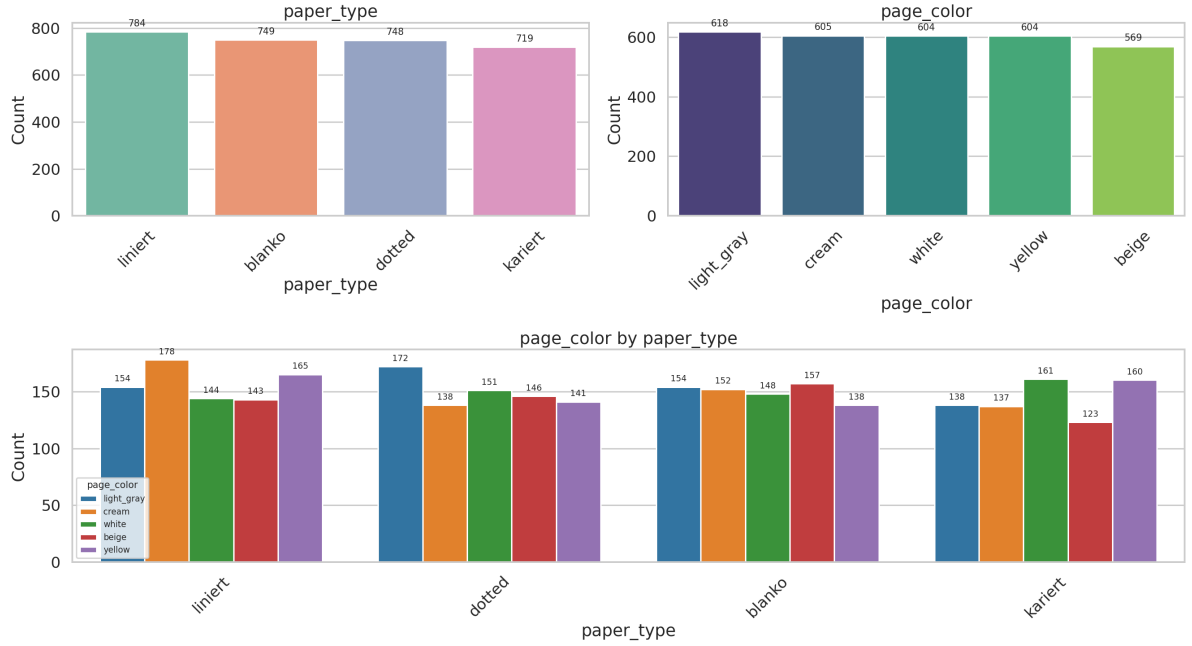


Figure 2.2.: Distribution of dynamically generated paper characteristics, including ruling types and paper colors, ensuring high visual background variance.

Handwriting Characteristics Simulation

A core component of the implementation is the simulation of realistic writing processes. Rather than rendering text statically, the system varies the visual output based on defined “writing styles” to simulate emotional or situational influences on the author:

- **“Stressed” mode:** Simulates hasty documentation under time pressure by enlarging the font, increasing the rotation and slope of the text block, and reducing line spacing.
- **“Careful” mode:** Represents conscientious entries characterized by a smaller font size, straight alignment, and wider line spacing.
- **“Normal” mode:** The baseline behavior representing standard, legible handwriting.

In addition to varying the pen color (black, blue), the system implements random transparency effects (alpha blending) to imitate the uneven flow of ink or fading pens over time.

2. Materials and Methods

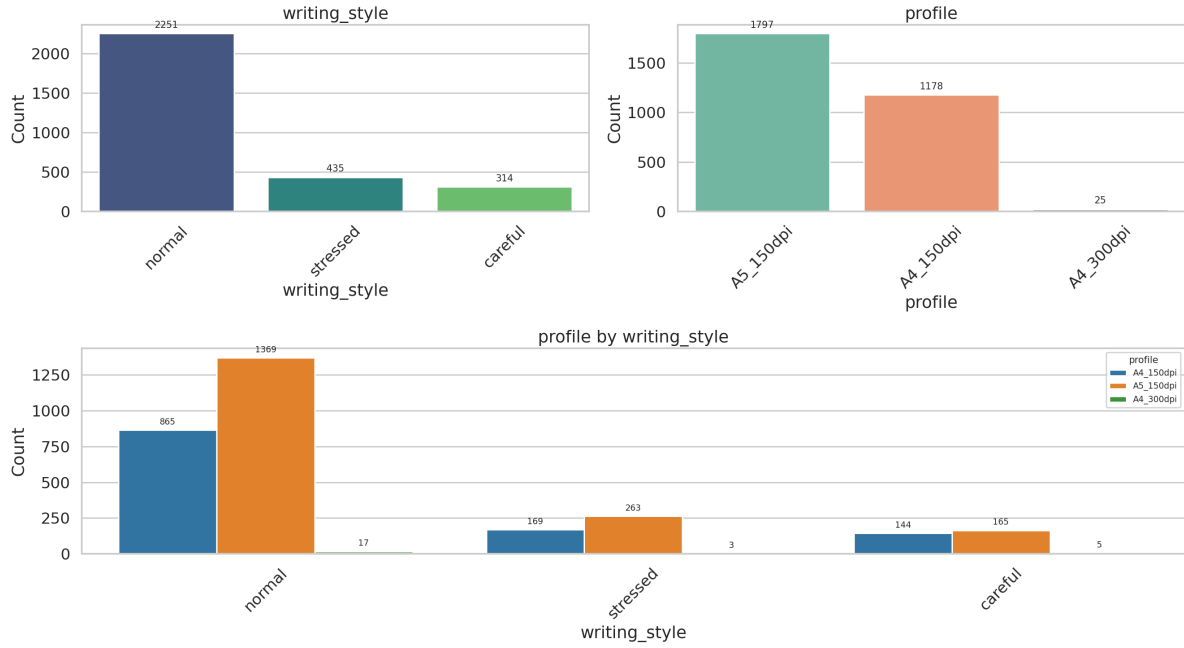


Figure 2.3.: Frequency of simulated writing styles mapped against the selected paper profiles, illustrating the variance in documentation behavior.

Technical Pipeline Architecture

To process the dataset efficiently, the software employs multiprocessing. A pool of worker processes generates image-text pairs in parallel, with each worker utilizing its own random seed to guarantee deterministic yet highly diverse results. Ultimately, the system exports the data in two formats:

- **Image data:** Synthetic documents (PNG/JPG) rendered in realistic resolutions.
- **Annotations:** Associated JSON files containing the ground truth text and relevant meta-data (scenario ID, font used, writing style) to enable supervised learning and evaluation.

2.3. Extraction Pipeline Architectures

The core objective of this thesis is to identify the most robust and accurate method for processing handwritten nursing notes in a robotic context. As illustrated in Figure 2.4, the Vision Routing Agent serves as a universal preprocessing stage to autonomously select the highest-quality frame. Subsequently, the information extraction is evaluated across three distinct, mutually exclusive architectural paradigms.

1. **Baseline Pipeline (OCR + LLM):** This traditional two-stage approach relies on explicit optical character recognition. The selected image is first processed by standard OCR frameworks (such as EasyOCR or Surya OCR) to extract raw, unstructured text. In the second stage, this raw text is passed to a text-only Large Language Model (e.g., ChatGPT or Mistral) in a Few-shot setting to correct recognition errors and extract structured semantic data. This pipeline serves as the benchmark for classical, explicit text extraction.
2. **VLM End-to-End Pipeline (OCR-Free):** Representing a paradigm shift in document understanding, this architecture completely bypasses the explicit OCR step. A multimodal

Vision Language Model (e.g., Qwen3-VL or Mistral-Vision) directly analyzes the raw image pixels and generates the structured semantic extraction and summarization in a single, unified inference step. This approach is evaluated to determine if joint visual-semantic reasoning can overcome the heuristic limitations of classical OCR in recognizing complex handwriting.

3. **Hybrid Pipeline (VLM-OCR + LLM):** This approach combines the strengths of both paradigms by utilizing a specialized vision model strictly for transcription prior to semantic processing. An advanced VLM tailored specifically for text extraction (such as LightOnOCR or GOT-OCR2.0) processes the image to produce highly accurate raw text. This transcription is then handed off to a powerful text-only LLM, which acts as a reasoning agent to summarize, structure, and refine the extracted information. This hybrid method aims to evaluate whether decoupling the visual perception task from complex semantic reasoning yields superior accuracy and control compared to a purely end-to-end architecture.

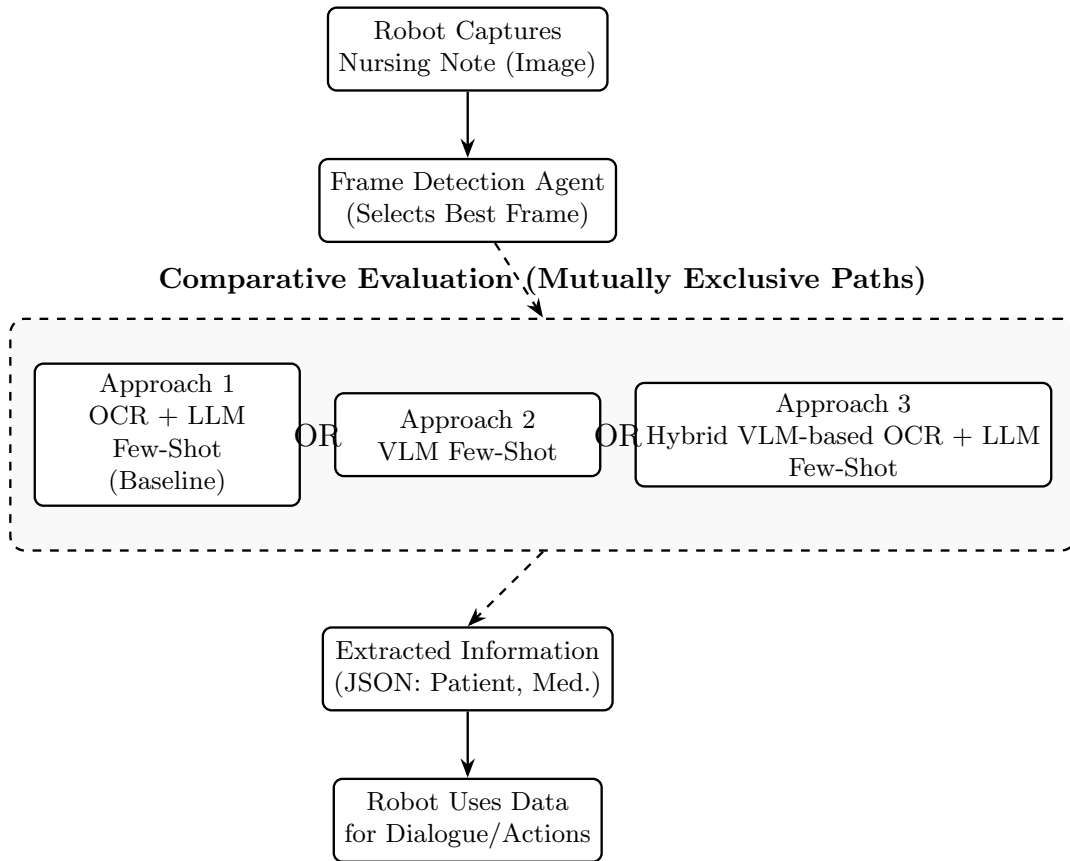


Figure 2.4.: Overview of the system architecture. The Vision Routing Agent selects the optimal frame, which is then passed into the evaluation module (dashed box). Within this module, **one of three mutually exclusive extraction pipelines** is executed. The resulting output is a standardized JSON.

By benchmarking these distinct pipelines against each other using the evaluation metrics defined in Section 2.6, this thesis empirically determines the optimal balance between computational efficiency (edge vs. cloud) and semantic accuracy for automated nursing documentation.

2.4. Model Selection and Baselines

2.4.1. Optical Character Recognition (The Text Extractors)

The evaluation compared three conceptually different open-source models:

2.4.1.1. EasyOCR (Deep Learning Approaches)

EasyOCR is a lightweight, open-source framework based on a classical two-stage deep learning pipeline for optical character recognition (39). It utilizes the CRAFT (Character Region Awareness for Text Detection) model to accurately localize text bounding boxes (18), followed by a CRNN (Convolutional Recurrent Neural Network) combined with a CTC (Connectionist Temporal Classification) decoder for the subsequent character recognition (19). In this thesis, EasyOCR was selected as the baseline model because of its highly resource-efficient architecture and its established role as a standard benchmark tool for traditional text extraction tasks.

FIGURE: EasyOCR PIPELINE ARCHITECTURE (CRAFT + CRNN-CTC)

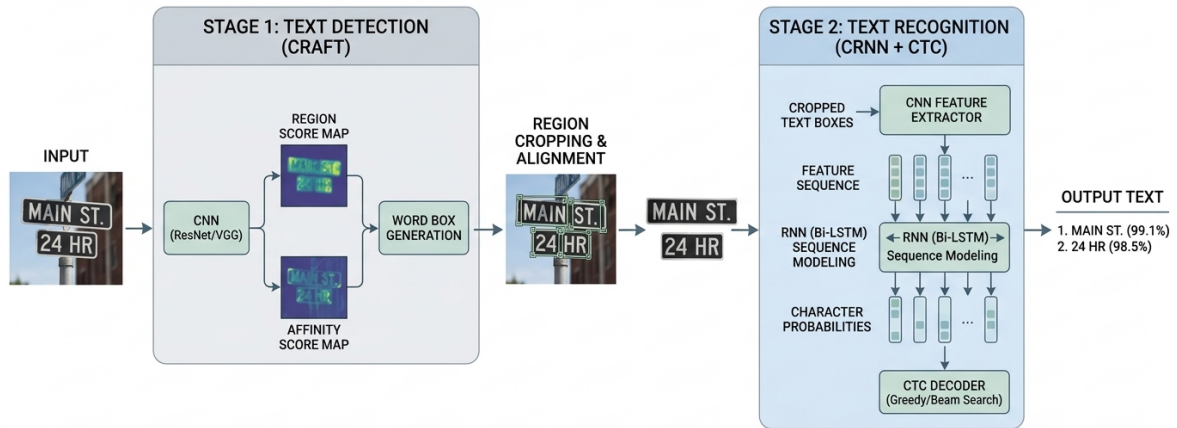


Figure 2.5.: Architecture of the EasyOCR pipeline, demonstrating the classical two-stage deep learning approach. In Stage 1, the CRAFT module generates region and affinity score maps to detect text bounding boxes. Following image cropping, Stage 2 utilizes a CRNN architecture with a CTC decoder to perform sequence-level text recognition on the isolated boxes (40).

2.4.1.2. Surya OCR (Transformer-based OCR)

Surya is a modern, highly complex open-source document OCR toolkit designed specifically for advanced document analysis, including layout parsing and reading order detection (21). In contrast to the traditional CNN/RNN pipeline of EasyOCR, Surya OCR is based on a modified encoder-decoder Transformer architecture inspired by the Donut (Document Understanding Transformer) model (20). By adopting Donut’s foundational approach—which maps raw document images directly to structured text without relying on brittle text-detection bounding boxes—Surya achieves a deeper semantic understanding of the document layout. It first processes the input image through a central foundation vision encoder to generate contextualized

visual feature representations, before a specialized language decoder sequentially predicts and extracts the characters.

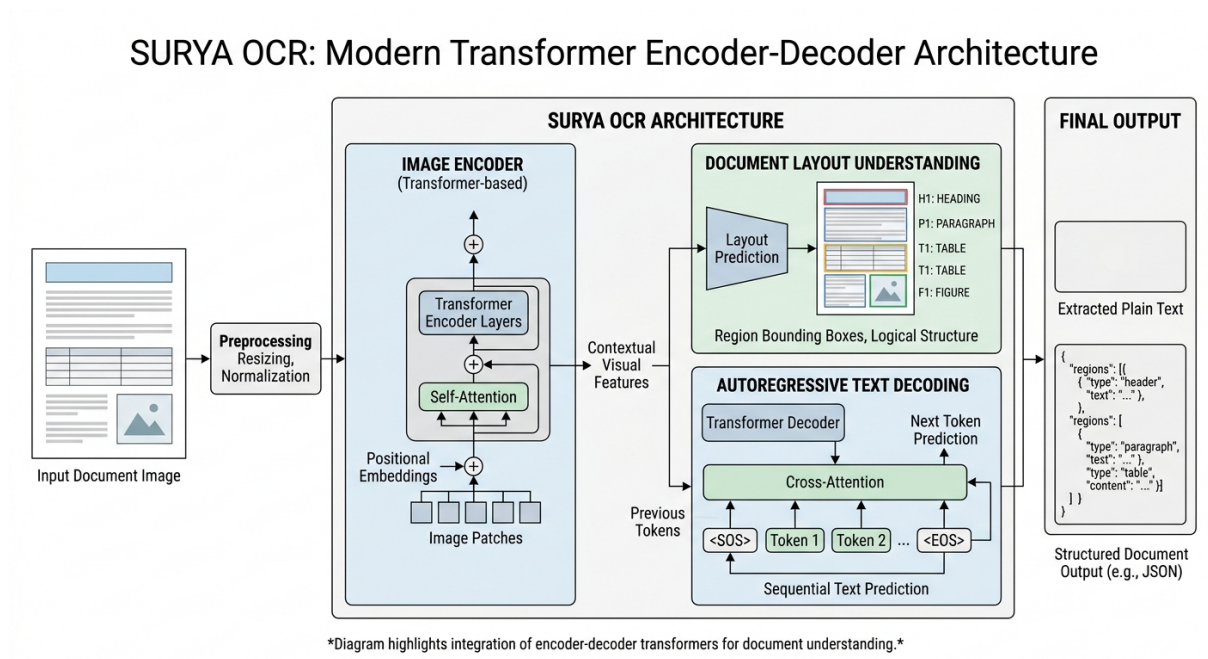


Figure 2.6.: Architecture of Surya OCR, illustrating the Transformer-based encoder-decoder pipeline. The image encoder processes document patches to extract contextual visual features, which are then utilized by the document layout module for structural parsing and the autoregressive language decoder for sequential text extraction (41).

2.4.1.3. LightOnOCR-2-1B (End-to-End VLM)

LightOnOCR-2 is a state-of-the-art, 1B-parameter multilingual Vision-Language Model (VLM) designed for end-to-end optical character recognition (42). Unlike traditional multi-stage pipelines that rely on separate detection and recognition modules, LightOnOCR-2 processes document images directly to generate clean, naturally ordered text. Architecturally, the model combines a vision encoder with a Mistral-based language model, allowing visual features to be integrated into a powerful autoregressive text generation framework (42). This design enables the model not only to transcribe text, but also to better handle complex layouts, multilingual content, and semantically structured documents. Beyond standard text extraction, LightOnOCR-2 is also extended to predict normalized bounding boxes for embedded images, making it a broader document understanding model rather than a pure OCR engine (42). In this thesis, it was selected as the VLM-based representative because it reflects the newest trend of using compact but highly capable generative vision-language architectures for OCR and document parsing.

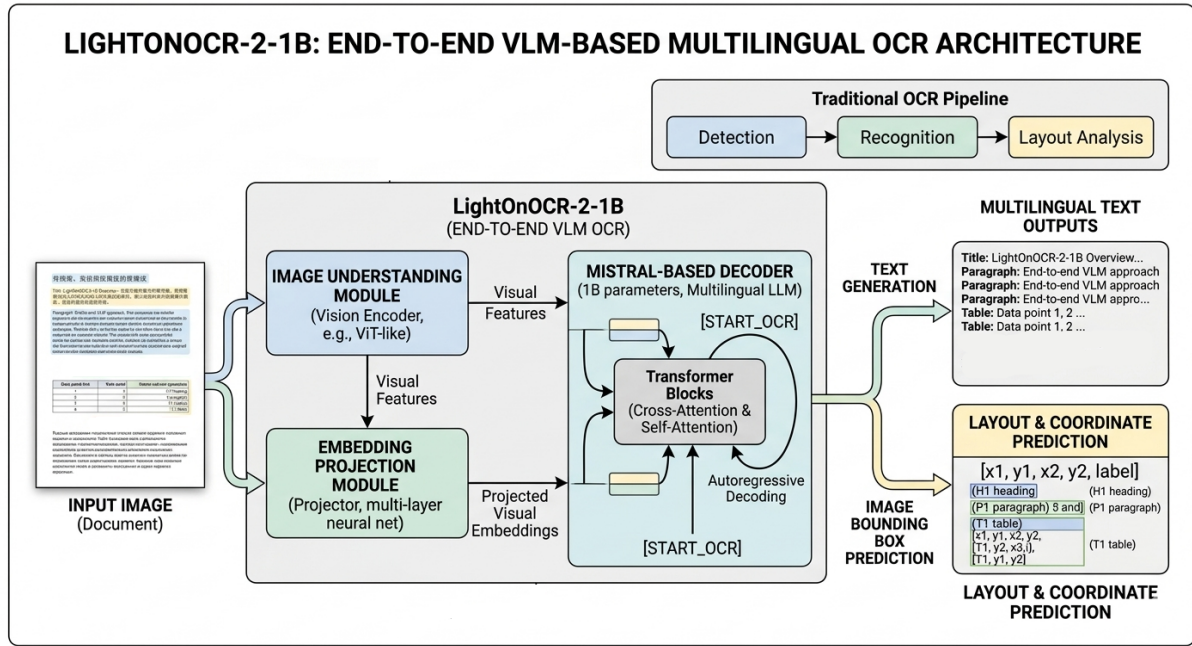


Figure 2.7.: End-to-end Vision-Language Model architecture of LightOnOCR-2-1B. Unlike traditional pipelines (top right), the VLM processes the input image through a vision encoder and an embedding projection layer. The resulting visual features are fed into a Mistral-based autoregressive decoder, which simultaneously generates multilingual text and predicts normalized bounding boxes for document layout understanding (43).

2.4.2. Large Language Models (The Reasoners)

In the disjointed extraction pipeline, Large Language Models (LLMs) act as reasoning and post-processing components. Their task is not to perform text recognition directly from the image, but rather to interpret noisy OCR output, resolve recognition ambiguities, and transform the extracted text into the required structured format. In this sense, the LLMs operate as semantic correction layers placed after the OCR stage.

To ensure deterministic behavior and full reproducibility of the information extraction results, strictly constrained decoding parameters were applied across all Large Language Models (LLMs). The generation temperature was set to zero (`temperature=0.0`), and probabilistic sampling was disabled (`do_sample=False`). This greedy decoding strategy forces the models to consistently output the most probable token sequence, eliminating random variations between different evaluation runs. By removing generative randomness, the models' raw error-correction capabilities and structural understanding of the noisy OCR inputs can be evaluated objectively and reliably.

All LLMs/VLMs were evaluated under the same prompting logic and extraction objective. This design ensures that the comparison focuses on each model's ability to recover meaning from imperfect OCR text, preserve medically relevant information, and generate consistent structured outputs from noisy document content.

2.4.2.1. Qwen3-8B-instruction (4-bit)

Qwen3-8B-Instruct is an open-weight large language model developed by Alibaba Cloud (44). It features approximately 8 billion parameters and is built upon a standard dense Transformer architecture optimized with grouped-query attention (GQA) for efficient inference. The model was fine-tuned specifically for instruction following and complex reasoning tasks. For this thesis, the model weights were sourced from the Hugging Face model hub. To accommodate VRAM constraints without significantly degrading performance, the model was loaded using 4-bit quantization via the `bitsandbytes` library using the NF4 data type (45). This allows the model to run efficiently on consumer-grade GPUs while maintaining its extensive context window capabilities.

2.4.2.2. Chatgpt4o-mini

ChatGPT-4o-mini is a proprietary, closed-source foundation model developed by OpenAI (46). Designed as a highly efficient and cost-effective successor to previous lightweight models, it boasts a 128,000-token context window. Because the model weights are not publicly available, it was accessed exclusively via OpenAI’s REST API using the `gpt-4o-mini` endpoint. In this text-only stage of the pipeline, the model was fed the raw OCR strings through standard system and user message payloads. Its inclusion provides a benchmark representing state-of-the-art commercial instruction tuning—which has proven highly effective in medical literature extraction (47)—against which the open-weight local models are compared.

2.4.2.3. Ministral-8B

Ministral-8B is an open-weight language model developed by Mistral AI, explicitly engineered for edge computing and high-efficiency reasoning (48). It contains roughly 8 billion parameters and utilizes an innovative sliding-window attention mechanism to efficiently process long context sequences. While the model weights are available openly, for the purposes of this study, Ministral-8B was accessed via Mistral AI’s official REST API (`ministral-8b-latest` endpoint) to maintain consistent inference speed and environment parity with the other cloud models. The model was fed the raw OCR strings through standard system and user message payloads. Unlike larger foundational models, Ministral-8B is heavily optimized for prompt adherence and structured data extraction, making it a highly relevant baseline for clinical text processing.

2.4.3. Vision Language Models (The End-to-End Extractors)

Unlike the text-only LLMs, Vision Language Models (VLMs) do not rely on a prior OCR step. Instead, they ingest the raw document image alongside a text prompt, relying on an integrated vision encoder to tokenize the image into a format the language model can process.

2.4.3.1. Qwen3-VL-8B-instruction (4-bit)

Qwen3-VL-8B-Instruct is the multimodal counterpart to the Qwen3 text model, developed by Alibaba Cloud (49). It integrates the 8-billion-parameter language model with an advanced Vision Transformer (ViT). A defining technical feature of this model is its dynamic resolution processing; instead of resizing all images to a fixed grid, it dynamically scales visual tokens based on the original aspect ratio and resolution of the input document (49). The weights were accessed via Hugging Face and loaded in 4-bit precision (45). This allows the model to process high-resolution clinical documents—where fine details like checkboxes and small handwritten notes are critical—within a limited VRAM footprint.

2.4.3.2. Chatgpt4o-mini

For the multimodal pipeline, the same ChatGPT-4o-mini model by OpenAI was utilized (46), but invoked using its vision capabilities. Accessed via the API, the document images were encoded as Base64 strings and passed directly into the message payload alongside the extraction instructions. OpenAI’s architecture natively processes these visual tokens alongside the text prompt, bypassing traditional OCR completely. This setup represents the highest tier of cloud-based visual document understanding and serves as the primary proprietary baseline for end-to-end extraction.

2.4.3.3. Pixtral-12B

Pixtral-12B is a multimodal foundational model developed by Mistral AI (50). Built upon a 12-billion-parameter language base, it incorporates a natively integrated vision encoder trained from scratch, capable of processing images of arbitrary sizes without aggressive down-sampling. Unlike the locally deployed Qwen models, Pixtral-12B was accessed via Mistral AI’s official REST API (pixtral-12b-2410 endpoint). The document images were encoded as Base64 strings and transmitted in the API payload alongside the extraction prompts. Pixtral-12B is particularly notable for its ability to seamlessly interleave image and text tokens within its 128,000-token context window (50). Its inclusion allows for an evaluation of how a medium-scale (12B) VLM compares against both local 8B counterparts and OpenAI’s commercial APIs when analyzing dense clinical forms.

2.4.4. Prompt Engineering and Iterative Refinement

In the course of the evaluation, the prompting strategy was iteratively refined to optimize the Vision-Language Models’ extraction performance. Initially, the prompt (Version 1) instructed the models to first generate a full verbatim transcription of the handwritten notes before extracting features. However, empirical observations indicated that forcing a strict transcription step often led to unnecessary verbosity and sometimes propagated character-level hallucinations into the final summary.

Consequently, the prompt was updated to a purely end-to-end approach (Version 2). This optimized prompt explicitly omits the transcription step, instructing the model to directly synthesize a clinical summary and extract key medical features directly from the visual input. This adjustment significantly streamlined the output and focused the model’s attention on semantic comprehension rather than pure character recognition. The full text of both prompt iterations is provided in Appendix A.1.3.

2.5. Edge-AI Frame Selection (Vision Routing Agent)

One of the biggest challenges in service robotics is variable image quality due to motion blur and changing lighting conditions. While classic algorithmic filters, such as the variance of the Laplacian (51), can efficiently compute mathematical image sharpness, they lack a semantic understanding of the visual context. Consequently, they cannot distinguish between a perfectly focused but irrelevant background and a slightly blurred, yet readable, medical document.

To overcome this limitation, a *Vision Routing Agent* was placed upstream of the actual extraction pipeline. This approach is based on the *Cascade Architecture* (Small-to-Large) paradigm, which has recently emerged as a highly efficient strategy in multimodal systems (52; 53). By

utilizing a lightweight edge model to rapidly pre-filter and route incoming frames based on semantic relevance, the system ensures that only valid, document-containing images went in the pipeline.

2.5.1. Lightweight VLM and 4-Bit Quantization

Instead of wasting resource-intensive API calls or expensive inference time on faulty images, a compact vision language model (Qwen3-VL-4B-Instruct) evaluates the robot’s incoming camera frames in real time (54).

To run this model locally with minimal latency on a mid-range edge device (e.g., an RTX 3060 with 12 GB VRAM), it is loaded using 4-bit quantization (55). This heavily reduces the peak VRAM usage, allowing the model to fit comfortably into restricted memory environments without sacrificing reasoning capabilities.

Additionally, the high-resolution image from the robot’s camera is dynamically downsampled by limiting the `max_pixels` parameter to approximately 500,000 pixels. This prevents VRAM overflow and intentionally biases the routing model towards global quality cues (e.g., strong motion blur or missing document edges) rather than processing unnecessary local texture details.

2.5.2. Structured Few-Shot Prompting and Routing Logic

To ensure deterministic and parseable behavior from the Vision Language Model, the routing agent employs a structured system prompt utilizing *few-shot prompting* techniques (23). This prompt enforces a precise evaluation of image quality (assigning a score from 0 to 100) based on strict deduction rules regarding motion blur, rotation, and incomplete page borders. Instead of unstructured free text, the agent is constrained to output its continuous frame analysis in a structured JSON format (e.g., `{"score": 85, "reason": "sharp and fully visible"}`), a practice crucial for integrating generative models into automated programmatic pipelines (56).

To guarantee absolute system stability in the event of model hallucination or malformed JSON outputs, a robust two-stage parsing fallback mechanism was implemented. The system first attempts to extract the JSON object via regular expressions (Regex). If the model fails to generate valid JSON architecture entirely, a secondary Regex fallback isolates the numerical score directly from the raw string. This ensures the routing logic never crashes during live HRI operations.

Furthermore, generative randomness was eliminated by applying greedy decoding (`do_sample=False`) and strictly limiting the output length (`max_new_tokens=60`). This forces the model to respond as quickly and predictably as possible. The entire evaluation pipeline is executed headlessly, allowing the agent to function seamlessly as an invisible background service on the robotic hardware.

Based on a predefined threshold (e.g., `ACCEPT_THRESHOLD = 70`), the routing logic makes real-time decisions:

- **Discard (Score < 70):** Frames that suffer from severe blurring, insufficient lighting, clipped document edges, or that do not display a relevant nursing note are immediately discarded locally at the edge.

- **Forward (Score ≥ 70):** As soon as a usable frame satisfies both the semantic and optical threshold, the video stream processing is halted. Only this identified *Best Frame* is forwarded to the computationally intensive extraction pipeline.

By preventing invalid frames from reaching the extraction stage, the downstream pipeline (which utilizes larger 8B/14B models or APIs) can operate efficiently without causing memory overflows, generating hallucinated data, or bottlenecking the overall system.

2.5.3. Dynamic Frame Sampling, Batching

Initial implementations of the Vision Routing Agent captured a continuous stream of images from the robot’s camera and processed them sequentially. In the context of Human-Robot Interaction (HRI), executing individual forward passes for every single frame leads to prolonged cumulative latency, which disrupts the natural conversational flow and creates unacceptable delays.

To overcome this bottleneck, a two-step optimization protocol was designed, combining *data reduction* with *dynamic GPU batching*:

1. **Window-Based Frame Reduction:** Instead of processing the entire raw video stream, the system intentionally drops redundant or highly degraded frames before they reach the model. The input is thereby heavily decimated, leaving a highly representative subset of exactly 8 images for processing.
2. **Parallel Batch Inference:** Rather than evaluating these remaining 8 frames sequentially, the system loads them simultaneously into RAM using a defined `batch_size`. The vision processor packs the images and their corresponding prompts into a padded tensor. The local GPU then executes highly parallelized forward passes, processing multiple frames simultaneously to maximize hardware utilization.

Once the batches are scored, the agent evaluates the results. If at least one frame reaches or exceeds the predefined quality threshold (e.g., `ACCEPT_THRESHOLD = 70`), the system selects the frame with the highest relative score and forwards only this optimal image to the computationally intensive extraction pipeline. If no frame meets the criteria, the robot verbally requests the user to hold the document more steadily.

2.5.4. In-Memory Processing and Data Privacy (GDPR Compliance)

For the operational deployment of the *Vision Routing Agent* on the robot, writing the captured camera frames to persistent storage (Disk I/O) was deliberately avoided. Instead of physically saving a batch of images to a hard drive before loading them into the model for evaluation, the system processes the live video stream directly via *in-memory processing*.

Each captured frame is held strictly as a volatile array within the system’s Random Access Memory (RAM) and is passed to the Qwen3-VL model without ever being written to disk. If a frame in the batch reaches the `ACCEPT_THRESHOLD`, the loop is terminated, and any remaining unprocessed images in the volatile memory are immediately discarded.

This architectural approach, rooted in the principle of *Privacy by Design*, offers two significant advantages for service robotics in the healthcare sector (57):

1. **Reduced Latency:** By eliminating disk read and write operations, the system avoids common I/O bottlenecks. Maintaining the image tensors exclusively in RAM ensures that the dynamic GPU batching (as described in Section 2.5.3) can operate at maximum efficiency without being slowed down by storage hardware limitations.
2. **Data Privacy and GDPR Compliance:** Nursing notes inherently contain highly sensitive, personally identifiable health information (e.g., patient names, room numbers, diagnoses). By utilizing strict in-memory processing, the system minimizes the risk of data leaks and complies with the data minimization principles of the General Data Protection Regulation (GDPR) (58). Flawed, blurry, or discarded images are instantly cleared from the volatile memory. Only the successfully validated *Best Frame* is temporarily maintained in RAM for the downstream extraction pipeline, ensuring that no permanent digital copies of sensitive medical documents are ever left behind on the robot’s local hardware.

2.6. Human-in-the-Loop and Agentic Workflow

Due to the hardware limitations of the robotic platform, an experimental Python-based proof-of-concept is developed to simulate real-world clinical deployment. Following the quantitative evaluation of the extraction architectures (OCR+LLM vs. end-to-end VLM), the best-performing and legally compliant system will be integrated into an interactive software environment. To maximize reliability and safety in sensitive clinical contexts, this system is designed to be implemented not as a purely autonomous, reactive pipeline, but rather as a *Human-in-the-Loop Agent*.

2.6.1. Agentic System Architecture

The proposed proof-of-concept leverages a modular, agent-based architecture to simulate the robotic interaction. The system pipeline consists of three primary components that sequentially process the input document and enable interactive verification:

- **Vision Routing Agent (Qwen3-VL-8B, 4-bit):** Serving as the primary perception module, this agent analyzes the raw input image of the document to determine its context and layout. It acts as a router that preprocesses the visual information and orchestrates the subsequent extraction process. A highly quantized 4-bit version of Qwen3-VL-8B is utilized here to maintain a minimal memory footprint while ensuring rapid visual comprehension.
- **Primary Extraction Model (Best-Performing Evaluated Model):** This represents the core analytical engine of the pipeline. Based on the evaluation, the most accurate and legally compliant model is slotted into this position. Its responsibility is to execute the high-precision extraction of medical terminology and structure the data into the final JSON format based on the router’s input.
- **Text-to-Speech (TTS) Agent (Qwen-TTS-Base-1.7):** To facilitate the *Human-in-the-Loop* interaction, this component translates the system’s extraction results, uncertainties, or queries into spoken language. This auditory feedback simulates the robotic interface, allowing to seamlessly verify the extracted clinical data or respond to the agent’s clarifying questions without needing to constantly monitor a screen.

2.6.2. Human-in-the-Loop Design

The core operational principle of this agentic workflow is to establish a collaborative synergy between the robotic system and the nursing staff. Rather than treating document processing as a rigid, one-way pipeline that forces a final output regardless of input quality, the system is designed to act as an interactive assistant. By prioritizing transparency and active error prevention, the workflow systematically incorporates human expertise to resolve ambiguities. Whenever the agent encounters challenging handwritten passages or contextually unclear clinical data, it shifts from autonomous processing to a cooperative dialogue, ensuring that critical medical information is verified by a human professional before being finalized.

To maximize operational safety and user trust in a clinical setting, this interactive workflow is governed by three primary states:

- **Status Updates and Time Bridging:** Given the inherent inference latency of large AI models, maintaining continuous communication with the user is critical. The system acts as a transparent intermediary, acoustically informing the nursing staff about ongoing background processes. By verbalizing intermediate steps (e.g., *“I have captured the document,”* or *“I am currently extracting the medication list”*), the agent bridges processing times and prevents the user from assuming a system failure during complex multimodal inference.
- **Interactive Clarification (Ask_Human):** If the model detects structural ambiguities, illegible handwriting, or medically implausible values (e.g., an unusually high dosage such as *“500 units of insulin”*), the agent actively halts the final extraction process. It formulates a highly targeted clarifying question, which is verbalized via the Text-to-Speech module (e.g., *“I cannot clearly decipher the second medication. Did you mean Novamin?”*). The clinician’s spoken response is subsequently fed back into the agent’s context window, allowing the model to resolve the uncertainty and safely complete the structured data entry without hallucinating.
- **Autonomous Extraction (Success):** If the document is clearly legible and the extracted entities exhibit high confidence and clinical plausibility, the model directly finalizes the structured JSON documentation. The agent subsequently concludes the workflow by providing a brief, acoustic confirmation to the user, verifying that the record has been successfully processed and saved.

By integrating this self-reflective *Human-in-the-Loop* mechanism, the system shifts from a rigid, error-prone extraction pipeline to a collaborative robotic assistant. This approach directly mitigates the high variance observed in zero-shot VLM evaluations by leveraging human expertise strictly when necessary, thereby ensuring maximum data integrity.

2.7. Metrics for Evaluation

This thesis evaluates two complementary capabilities required for automatic processing of handwritten nursing notes. Task 1 assesses transcription performance (image $\rightarrow$ text). Task 2 evaluates downstream understanding (summarization, question answering, and—where applicable—structured information extraction) based on transcribed text or directly from images (OCR+LLM vs. VLM).

2.7.1. Task 1: Transcription Performance

Given an input image, each method outputs a transcription hypothesis H , which is compared to the ground-truth reference transcription R . To ensure comparability across methods, R and H are normalized using a fixed policy (lowercasing and whitespace normalization; line breaks mapped to spaces; German umlauts, ß, punctuation, and numbers are preserved). Metrics are computed per sample and aggregated over the fixed test set.

Character Error Rate (CER): Character Error Rate (CER) measures transcription quality at the character level by computing the edit distance between the hypothesis H and the ground-truth reference R . A lower CER indicates a better transcription quality (59; 60).

$$\text{CER} = \frac{S + D + I}{N_{\text{char}}} \quad (2.1)$$

This formula represents the CER, where:

- S is the number of character substitutions,
- D is the number of character deletions,
- I is the number of character insertions,
- N_{char} is the number of characters in the reference transcription R .

Word Error Rate (WER): Word Error Rate (WER) evaluates transcription quality at the word level. It is computed analogously to CER, but the edit operations are applied to a sequence of whitespace-tokenized words rather than individual characters. A lower WER indicates better recognition of words and word order (61; 62).

$$\text{WER} = \frac{S + D + I}{N_{\text{word}}} \quad (2.2)$$

This formula represents the WER, where:

- S is the number of word substitutions,
- D is the number of word deletions,
- I is the number of word insertions,
- N_{word} is the number of words in the reference transcription R .

Normalized Edit Distance (NED): Normalized Edit Distance (NED) expresses the Levenshtein edit distance between R and H normalized by the length of the longer string. This yields a bounded measure that is more comparable across samples of different lengths. A lower NED indicates higher similarity (59).

$$\text{NED} = \frac{d_{\text{lev}}(R, H)}{\max(|R|, |H|)} \quad (2.3)$$

This formula represents NED, where:

2. Materials and Methods

- $d_{\text{lev}}(R, H)$ is the Levenshtein distance between the reference R and hypothesis H ,
- $|R|$ is the length of R in characters,
- $|H|$ is the length of H in characters.

Vocabulary overlap (unique-word Precision/Recall/F1): Vocabulary overlap evaluates whether the transcription covers a similar set of unique words as the ground truth. Let V_R and V_H denote the sets of unique words in R and H , respectively (after normalization and whitespace tokenization). The overlap is summarized using precision, recall, and F1-score (63); this metric is used as a supplementary indicator because it ignores word order and repeated occurrences.

$$\text{Precision}_{\text{vocab}} = \frac{|V_R \cap V_H|}{|V_H|} \quad (2.4)$$

$$\text{Recall}_{\text{vocab}} = \frac{|V_R \cap V_H|}{|V_R|} \quad (2.5)$$

$$\text{F1}_{\text{vocab}} = \frac{2 \cdot \text{Precision}_{\text{vocab}} \cdot \text{Recall}_{\text{vocab}}}{\text{Precision}_{\text{vocab}} + \text{Recall}_{\text{vocab}}} \quad (2.6)$$

(When both precision and recall are 0, F1 is defined as 0.)

This set of formulas represents vocabulary overlap, where:

- V_R is the set of unique words in the reference R ,
- V_H is the set of unique words in the hypothesis H ,
- $|\cdot|$ denotes set cardinality.

Sequence-based Accuracy (string similarity): Sequence-based accuracy reports a similarity score between reference and hypothesis as a single value. In this thesis, it is computed using a sequence similarity function $\text{sim}(R, H)$ based on normalized edit distance and reported as a percentage (59). This metric is included as a supplementary indicator of overall string similarity in addition to edit-distance metrics such as CER and WER.

$$\text{Accuracy}_{\text{seq}} = \text{sim}(R, H) = 1 - \frac{d_{\text{lev}}(R, H)}{\max(|R|, |H|)} \quad (2.7)$$

This formula represents the sequence-based accuracy, where:

- $\text{sim}(R, H)$ is a sequence similarity function producing values in $[0, 1]$ (reported in percent),
- $d_{\text{lev}}(R, H)$ is the Levenshtein distance between R and H .

2.7.2. Task 2: Downstream Understanding (Summarization and Extraction)

The second task evaluates how well the model preserves and expresses clinically relevant information when producing textual outputs (such as summaries and answers) and, where applicable, structured outputs (JSON field extraction).

Given a reference text (or reference summary) R and a generated output H , metrics are calculated for each sample and aggregated over the test set.

ROUGE (ROUGE-1, ROUGE-2, ROUGE-L): ROUGE measures lexical overlap between H and R using n -gram matching (ROUGE- n) and longest common subsequence matching (ROUGE-L) (64).

Let $G_n(R)$ and $G_n(H)$ denote the sets of unique n -grams in R and H , respectively, and let $\text{count}_R(g)$ and $\text{count}_H(g)$ denote the number of occurrences of n -gram g in R and H . The clipped n -gram overlap is:

$$\text{overlap}_n(R, H) = \sum_{g \in G_n(R) \cup G_n(H)} \min(\text{count}_H(g), \text{count}_R(g)). \quad (2.8)$$

ROUGE- n precision, recall, and F1 are then:

$$P_n = \frac{\text{overlap}_n(R, H)}{\sum_{g \in G_n(H)} \text{count}_H(g)}, \quad R_n = \frac{\text{overlap}_n(R, H)}{\sum_{g \in G_n(R)} \text{count}_R(g)}, \quad (2.9)$$

$$F1_n = \frac{2P_n R_n}{P_n + R_n}. \quad (2.10)$$

(When both P_n and R_n are 0, $F1_n$ is defined as 0.) For ROUGE-L, let $\text{LCS}(R, H)$ denote the length of the longest common subsequence between R and H (computed at the token level), and let $|T(R)|$ and $|T(H)|$ denote the number of tokens in R and H . ROUGE-L is defined analogously:

$$P_L = \frac{\text{LCS}(R, H)}{|T(H)|}, \quad R_L = \frac{\text{LCS}(R, H)}{|T(R)|}, \quad F1_L = \frac{2P_L R_L}{P_L + R_L}. \quad (2.11)$$

(When both P_L and R_L are 0, $F1_L$ is defined as 0.)

Medical term preservation (keyword-set Precision/Recall/F1): To specifically assess whether clinically important terminology is retained in LLM-generated summaries, we compute a term-preservation score based on domain-specific keyword matching (65).

Let $M(X)$ denote the set of medical/nursing keywords detected in text X (case-insensitive matching against a predefined German keyword list covering vital signs, medications, symptoms, and care activities).

Let $M_R = M(R)$ and $M_H = M(H)$, and define the overlap $k = |M_R \cap M_H|$.

Term preservation is quantified via precision, recall, and F1:

$$\text{Recall}_{\text{med}} = \frac{k}{|M_R|}, \quad \text{Precision}_{\text{med}} = \frac{k}{|M_H|}, \quad (2.12)$$

$$F1_{\text{med}} = \frac{2 \cdot \text{Precision}_{\text{med}} \cdot \text{Recall}_{\text{med}}}{\text{Precision}_{\text{med}} + \text{Recall}_{\text{med}}}. \quad (2.13)$$

Edge cases: If $|M_R| = 0$ (no reference medical terms), recall is defined as 1; if $|M_H| = 0$ (no hypothesis medical terms), precision is defined as 1 if $|M_R| = 0$ and 0 otherwise. When both precision and recall are 0, F1 is defined as 0.

BERTScore: While traditional metrics like the Character Error Rate (CER), ROUGE or Medical Term Preservation (F1) focus on exact lexical matching, they often fail to capture the true semantic accuracy of text generated by modern Large Language Models (LLMs) and Vision Language Models (VLMs).

These conventional metrics are grounded in Levenshtein distance or strict n -gram overlap, meaning they treat semantically plausible normalizations as insertion or substitution errors,

fundamentally misrepresenting a generative model’s comprehension capabilities (66).

For instance, if a VLM correctly extracts the semantic meaning of a handwritten medical note but uses a synonymous term or resolves an abbreviation (e.g., extracting “Ibuprofen” instead of the handwritten “Ibu”), n-gram-based metrics strictly penalize this output (66; 67). Furthermore, in summarization tasks, relying solely on a single human reference summary exacerbates this issue by penalizing semantically correct models that simply choose different, yet equally valid, phrasing.

To address these limitations, this thesis employs BERTScore (68). BERTScore is an automatic evaluation metric that estimates the semantic similarity between a candidate text and a reference text using contextualized token embeddings from a pre-trained Transformer language model (e.g., BERT or RoBERTa). Because it relies on deep semantic representations rather than rigid string matching, BERTScore is significantly less sensitive to purely lexical variations, making it highly effective for evaluating generative models in complex domains like healthcare.

To mathematically compute the BERTScore, the evaluation process is divided into three consecutive steps (68):

1. **Contextual Embeddings:** Let $x = \langle x_1, \dots, x_k \rangle$ represent the sequence of tokens in the ground-truth reference text, and $\hat{x} = \langle \hat{x}_1, \dots, \hat{x}_l \rangle$ represent the candidate token sequence generated by the model. Both sequences are passed through a pre-trained transformer model to extract a sequence of contextualized, high-dimensional vector embeddings, denoted as $\mathbf{x}_i$ for the reference and $\hat{\mathbf{x}}_j$ for the candidate.
2. **Pairwise Cosine Similarity:** To quantify the semantic equivalence between any reference token x_i and candidate token $\hat{x}_j$, the metric computes the cosine similarity between their respective embedding vectors. Assuming the vectors are pre-normalized to a length of 1, the cosine similarity simplifies to the inner product (dot product) $\mathbf{x}_i^\top \hat{\mathbf{x}}_j$.
3. **Maximum Similarity Matching and Scoring:** Unlike traditional metrics that require exact positional alignment, BERTScore pairs tokens based on maximum semantic similarity. For each token in one sequence, the algorithm searches the entire opposing sequence to find the single token with the highest cosine similarity (a process referred to as greedy matching in the original paper).

The *Recall* (R_{BERT}) measures how much information from the reference is successfully captured by the candidate text:

$$R_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} \mathbf{x}_i^\top \hat{\mathbf{x}}_j \quad (2.14)$$

Conversely, the *Precision* (P_{BERT}) measures how much of the generated candidate text is actually relevant to the reference:

$$P_{\text{BERT}} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} \mathbf{x}_i^\top \hat{\mathbf{x}}_j \quad (2.15)$$

Finally, to provide a single, balanced metric for the extraction quality, the F_{BERT} score is calculated as the harmonic mean of precision and recall:

$$F_{\text{BERT}} = 2 \frac{P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}} \quad (2.16)$$

To fully leverage this mathematical framework for the specific challenge of medical summarization, this thesis utilizes BERTScore in a *Document-as-Reference* (DocAsRef) approach (69). Rather than comparing the model output exclusively to a rigid human-written summary, the generated text is evaluated directly against the full, unsummarized nursing notes (**notes**). In this setup, the BERTScore Recall explicitly indicates how much of the original document’s critical information is preserved, while the Precision measures the factual consistency (absence of hallucinations). By shifting from exact sequence alignment to contextualized token embeddings, this approach accurately measures semantic completeness without the harsh penalization of domain-specific synonyms.

3. Experiments and Results

3.1. Frame Selection Performance and Latency Optimization

Before evaluating the downstream text extraction capabilities (Tasks 1 and 2), the operational efficiency of the upstream Vision Routing Agent (described in Section 2.5) was assessed. For real-world deployment in service robotics, low system latency is equally as critical as high extraction accuracy, as prolonged processing delays directly disrupt the Human-Robot Interaction (HRI) flow.

3.1.1. Latency Reduction through Dynamic Batching

Performance profiling was conducted on a local edge device (NVIDIA RTX 3060, 12 GB VRAM) to measure the impact of the proposed two-step optimization protocol. The initial naive implementation—processing a continuous stream of 18 images sequentially—resulted in an inference time of approximately 2.1 seconds per frame. Consequently, processing the entire sequence required nearly 38 seconds of pure inference time, an unacceptable delay for interactive robotic scenarios.

Following the implementation of the window-based frame reduction and parallel batch inference, system performance improved drastically. Empirical measurements demonstrated that processing the defined representative subset of 8 frames in a single batch takes a total of only 9.69 seconds. By parallelizing the forward passes, the *effective inference time per frame* was reduced from 2.1 seconds to roughly 1.2 seconds ($9.69\text{ s} \div 8$). Overall, this architectural adjustment reduced the routing latency by approximately 75 % (from 38 seconds to under 10 seconds), bringing it into a range that can be comfortably bridged using the robot’s auditory text-to-speech feedback.

3.1.2. VRAM Consumption and Edge Viability

In addition to latency, the memory footprint of the Vision Routing Agent was evaluated to ensure its viability on mid-range edge hardware. Deploying the Qwen3-VL-4B-Instruct model with 4-bit quantization and dynamically downscaling the 5-megapixel camera input to a `max_pixels` limit of 500,000 proved highly effective. During active inference (batch processing), peak VRAM usage was measured at approximately 3.5 to 4.2 GB.

This low memory footprint confirms that the agent successfully operates within strict hardware constraints, leaving sufficient residual VRAM on the 12 GB GPU to potentially host localized downstream extraction models without triggering out-of-memory (OOM) errors.

3.1.3. Qualitative Routing Effectiveness and Limitations

Beyond purely quantitative metrics, the semantic decision-making capability of the Vision Routing Agent was qualitatively evaluated. Figure 3.1 illustrates a typical batch sequence of eight frames processed by the quantized Qwen3-VL-4B-Instruct model, capturing a printed synthetic nursing note under varying physical conditions.

3.1. Frame Selection Performance and Latency Optimization

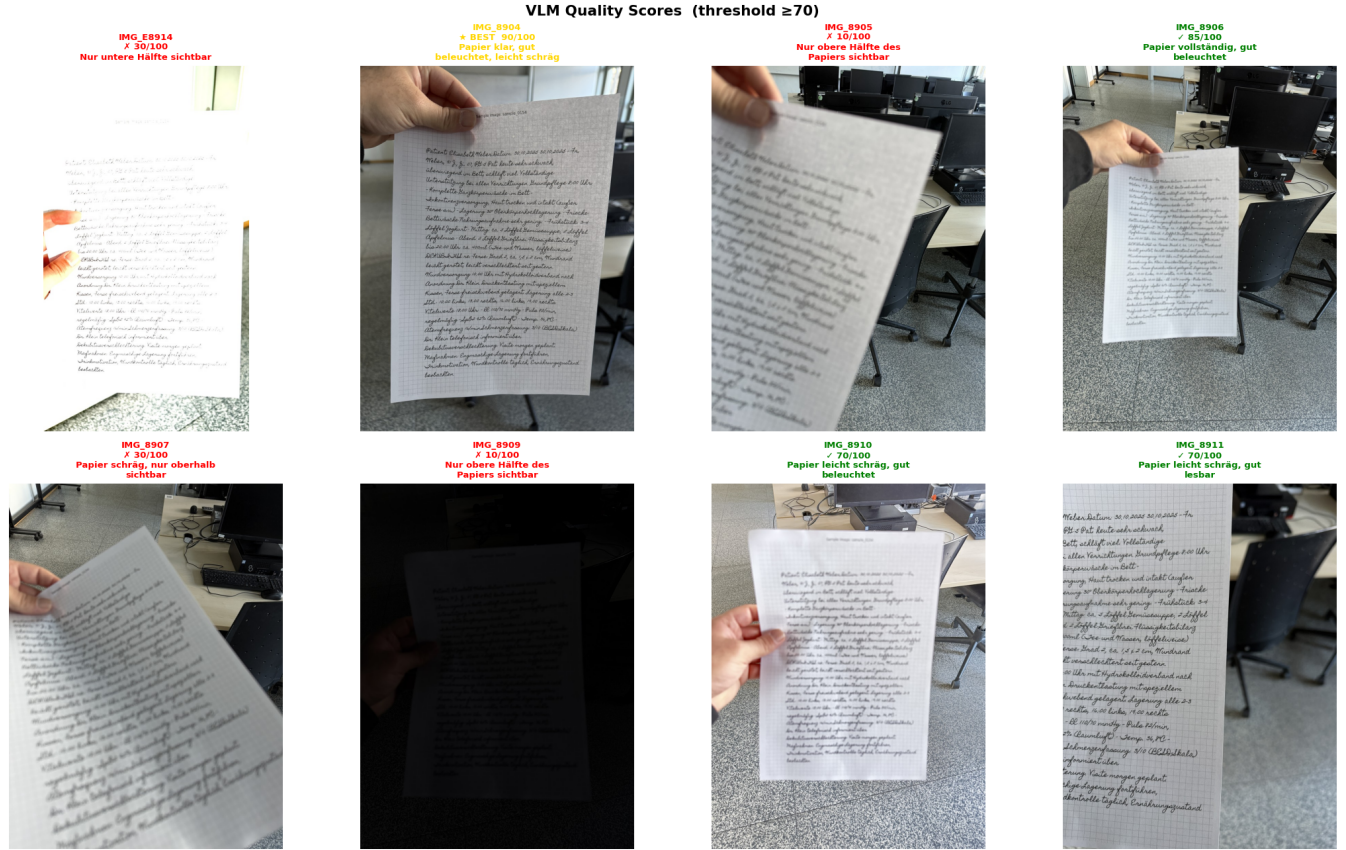


Figure 3.1.: Visualized scoring output of the Vision Routing Agent processing a batch of $N = 8$ consecutive frames capturing a printed synthetic nursing note. The model successfully identifies the optimal frame (yellow) and discards blurred or dark images (red), though it exhibits limitations in detecting abruptly cropped document boundaries (e.g., bottom right).

The agent successfully differentiates between severely degraded images and readable content. As dictated by the structured system prompt, the model correctly penalizes heavily blurred or underexposed frames (e.g., IMG.8905 and IMG.8909), assigning them non-passing scores of 10/100. Consequently, the agent correctly identifies IMG.8904 as the globally optimal "Best Frame" (score 90/100), accompanied by the valid reasoning "*Papier klar, gut beleuchtet, leicht schräg*" (paper clear, well-lit, slightly tilted). This qualitative output confirms that the upstream routing logic effectively shields the downstream extraction pipeline from the most severe forms of visual corruption, primarily motion blur and severe underexposure.

However, the evaluation also reveals notable limitations intrinsic to lightweight 4B-parameter Vision-Language Models operating at the edge. While the model correctly identifies the best overall frame, it occasionally struggles with spatial boundary awareness on extreme edge cases. For instance, the model assigns a passing score of 70/100 to IMG.8911, a frame where the document is physically cut in half by the camera boundary. The model incorrectly reasons that the paper is "*vollständig*" (complete) or "*gut lesbar*" (readable), presumably because the visible text lines remain sharply focused and mathematically detectable to the model's convolution layers.

3.2. Task 1: Transcription Performance (Image-to-Text)

This section presents the isolated results of the text recognition (Task 1). According to the model selection defined in Section 2.4.1, the performance of the classical OCR methods is first evaluated before they are used in the downstream pipeline architectures (Task 2).

3.2.0.1. Evaluation Dataset ($N = 559$)

To establish robust baselines for transcription performance, the pure OCR engines were evaluated on a comprehensive test split. This dataset (`test_set_559.json`) consists of 559 synthetic document images ($N = 559$), representing approximately 50% of the entire generated corpus.

This large evaluation dataset was rigorously stratified to prevent statistical bias during model assessment. As demonstrated in Figure 3.2, the dataset maintains an exceptionally balanced distribution across all critical visual parameters. The four paper ruling types (blank, lined, dotted, grid) and the five background colors are nearly equally represented. This ensures that the text extraction algorithms are rigorously tested against every possible combination of visual noise and background interference present in clinical reality.

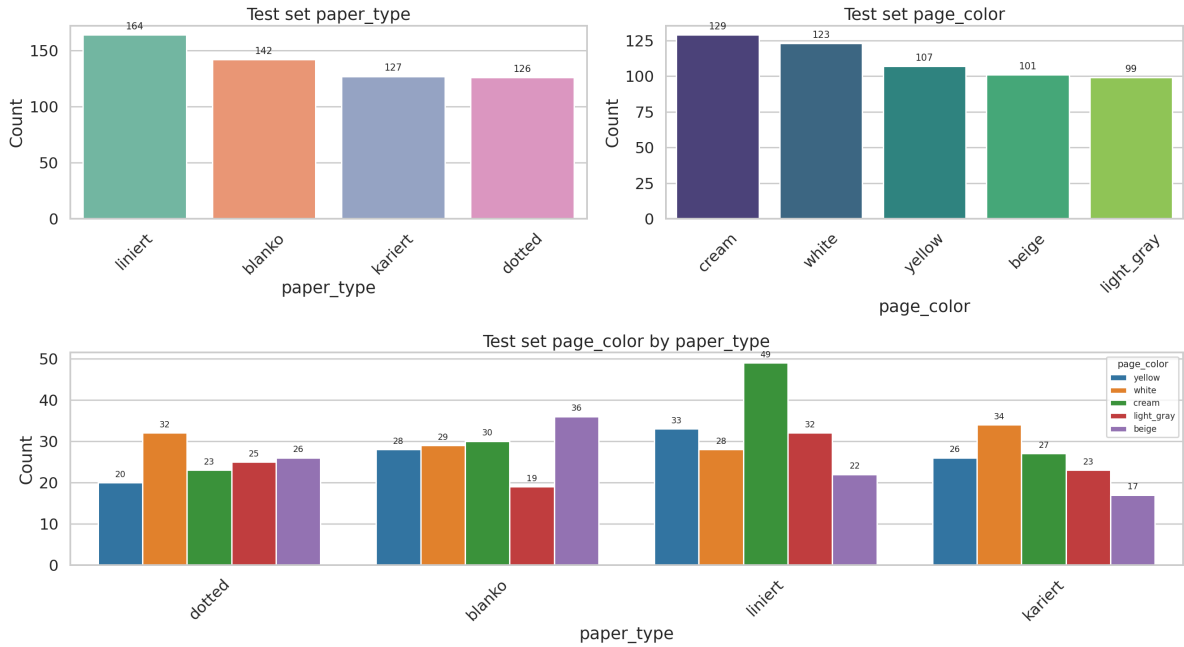


Figure 3.2.: Distribution of visual parameters (paper type and page color) within the $N = 559$ evaluation dataset, confirming the balanced stratification of the synthetic images used for OCR baseline testing.

3.2.1. Evaluation Of OCRs

3.2.1.1. Evaluation of EasyOCR

As the first baseline of the extraction pipeline, the EasyOCR model was evaluated on the stratified test dataset (`test_set_559.json`, $N = 559$). The quantitative results, based on the normalization guidelines defined in Section 2.7.1, are summarized in Table 3.1.

The analysis shows that isolated text recognition of complex handwritten nursing notes poses a significant challenge for this model. The average exact sequence accuracy is only 13.62 %

3.2. Task 1: Transcription Performance (Image-to-Text)

Table 3.1.: Results of the evaluation of EasyOCR on the synthetic test dataset ($N = 559$). All metrics are given as percentages (%). The definition of the metrics corresponds to section 2.7.1.

Metric	Mean	Median	Std. Dev.	Range
Accuracy	13,62	9,07	13,00	0,00 – 75,67
CER	60,67	65,49	19,60	12,53 – 100,00
WER	94,45	98,15	11,59	39,22 – 125,45
NED	60,66	65,49	19,60	12,53 – 100,00
Vocab Precision	12,88	6,67	14,73	0,00 – 65,31
Vocab Recall	13,06	6,25	15,49	0,00 – 68,09
Vocab F1-Score	12,92	6,11	15,10	0,00 – 66,67

(median: 9.07%). This low performance level is clearly underscored by the high error rates at the character and word levels: The mean Character Error Rate (CER) is 60.67 %, meaning that, on average, more than half of all characters per sample are transcribed incorrectly. The Word Error Rate (WER) is even more serious, averaging 94.45 %. A median WER of 98.15 % illustrates that in the vast majority of samples, almost no words were recognized completely without errors and segmented correctly.

The vocabulary metrics, which measure the pure overlap of unique words in the text regardless of their exact order, are also extremely low, with a mean F1 score of 12.92 %. The significant standard deviations (e. g., 19.60 % for CER) and the wide range of values indicate that while the model achieves moderate partial success with a few, presumably very legible samples, it does not provide reliable transcription on a broader scale.

These massive deficiencies in pure text recognition form a critical starting point for the subsequent evaluations in Task 2. In particular, they motivate the investigation of the question of whether downstream Large Language Models (LLMs) are able to extract correct semantic entities from such heavily noisy OCR results.

3.2.1.2. Evaluation of Surya OCR

Following the identical evaluation protocol on the same `test_set_559.json` dataset, Surya OCR was evaluated as the second text extractor. The results, summarized in Table 3.2, demonstrate a substantial performance leap compared to the EasyOCR baseline.

Surya OCR achieves a mean exact sequence accuracy of 62.96 %, with an even higher median of 72.58 %. This indicates that for the majority of the dataset, the model is highly capable of transcribing the handwritten nursing notes. The improvement is particularly striking regarding the Character Error Rate (CER), which drops to a mean of 14.43 %. The remarkable median CER of just 5.05 % proves that for more than half of the dataset, the transcription is almost character-perfect.

However, the Word Error Rate (WER) reveals an interesting characteristic of the model’s behavior on edge cases. While the median WER is solid at 27.27 % (a massive improvement over EasyOCR’s 98.15 %), the mean WER is pulled up to 43.72 % due to extreme outliers. The maximum WER of 1037.25 %, paired with a high standard deviation (69.39 %), suggests that

3. Experiments and Results

Table 3.2.: Results of the Surya OCR evaluation on the synthetic test dataset ($N = 559$). All metrics are expressed in percentages (%).

Metric	Mean	Median	Std. Dev.	Range
Accuracy	62.96	72.58	34.86	0.93 – 100.00
CER	14.43	5.05	23.79	0.00 – 292.78
WER	43.72	27.27	69.39	0.00 – 1037.25
NED	13.15	5.04	16.97	0.00 – 75.62
Vocab Precision	65.61	75.00	30.01	2.44 – 100.00
Vocab Recall	66.16	75.47	29.71	1.89 – 100.00
Vocab F1-Score	65.85	75.11	29.83	2.13 – 100.00

Surya OCR occasionally suffers from severe insertion errors or hallucination. On highly irregular or noisy samples, the model likely over-segments characters or generates significantly more words than present in the ground truth.

Despite these isolated outlier cases, the vocabulary metrics confirm strong semantic retention. With a mean Vocabulary F1-Score of 65.85 % and a median of 75.11 %, Surya OCR successfully captures the core terminology of the medical notes. Due to this vastly superior performance and its ability to preserve the underlying vocabulary, Surya OCR provides a much more robust textual foundation. Consequently, it is selected as the primary OCR engine for the OCR + LLM baseline pipeline (Approach 1) evaluated in Task 2.

3.2.1.3. Evaluation of LightOnOCR-2

To evaluate the potential of Vision-Language Models (VLMs) as primary text extractors, the LightOnOCR-2 model was tested on the same $N = 559$ valid samples. The results, presented in Table 3.3, reveal a fascinating dichotomy between typical performance and edge-case behavior, highlighting the generative nature of this architecture.

Table 3.3.: Results of the LightOnOCR-2 evaluation on the synthetic test dataset ($N = 559$). All metrics are expressed in percentages (%).

Metric	Mean	Median	Std. Dev.	Range
Accuracy	65.62	82.43	35.15	0.00 – 100.00
CER	40.21	4.44	139.11	0.00 – 1939.80
WER	67.14	21.31	203.14	0.00 – 2106.32
NED	17.16	4.44	23.84	0.00 – 96.84
Vocab Precision	69.32	80.00	29.13	0.00 – 100.00
Vocab Recall	71.57	80.85	27.84	0.00 – 100.00
Vocab F1-Score	70.10	80.00	28.43	0.00 – 100.00

On typical samples, LightOnOCR-2 demonstrates exceptional capabilities. It achieves a remarkable median exact sequence accuracy of 82.43 % and an outstanding median CER of just 4.44 %. Furthermore, its median Vocabulary F1-Score reaches 80.00 %, indicating a deep semantic understanding of the complex medical terminology present in the notes. For the vast majority of the dataset, LightOnOCR-2 provides the most accurate transcriptions among all

tested models.

However, the mean metrics and maximum error rates expose a critical vulnerability: extreme hallucination when encountering illegible or highly unstructured input. The mean CER is heavily skewed to 40.21 % (with a staggering maximum of 1939.80 %), and the mean WER spikes to 67.14 % (maximum 2106.32 %). These extreme outliers, accompanied by massive standard deviations (e.g., 203.14 % for WER), occur because the VLM attempts to generate plausible text even when the visual evidence is insufficient, leading to the creation of extensive text blocks that are entirely absent from the ground truth.

3.2.2. Evaluation of End-to-End VLMs

3.2.2.1. Evaluation Subset ($N = 48$)

Due to the significant computational overhead and API rate limits associated with multimodal inference, the end-to-end evaluation (for both transcription and downstream extraction) was conducted on a highly representative subset rather than the entire generated test corpus. This evaluation set was carefully assembled using a strict stratified sampling approach to ensure balanced coverage across all variation axes of the synthetic dataset, yielding a final set of 48 images ($N = 48$).

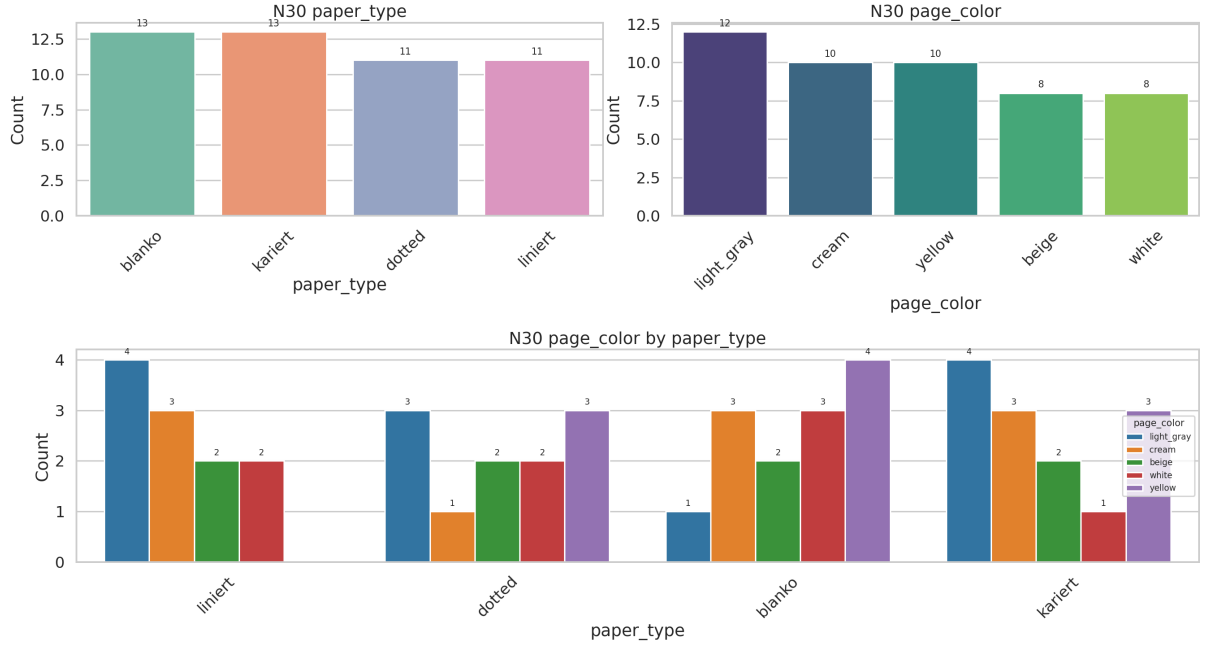


Figure 3.3.: Distribution of visual parameters (paper type and page color) within the $N = 48$ evaluation subset, demonstrating a balanced representation of the complete synthetic dataset’s variance.

To ensure statistical validity and a fair evaluation, this subset guarantees that all distinct features present in the original dataset are proportionally represented. Specifically, the resulting set achieves full font coverage (all 24 validated fonts are included at least once) and a text-length distribution closely matching the main synthetic corpus (approximately 15% short, 63% medium, and 23% long documents). To ensure full reproducibility and transparency, all 48 synthetic document images utilized in this evaluation subset are provided visually in Appendix

3. Experiments and Results

A.2.

Furthermore, as illustrated in Figure 3.3, the sampling strategy provides complete coverage of all four paper ruling types (blank, lined, dotted, grid) and all five page background colors (white, light grey, cream, beige, yellow). Additionally, all three handwriting styles (normal, careful, stressed) are represented, ensuring that the selected models are evaluated against the full spectrum of visual noise and formatting complexity.

3.2.2.2. Evaluation of Qwen3-VL-8B (4-bit)

Using this stratified subset, the quantized Qwen3-VL-8B (4-bit) model was tested for its ability to generate raw text from complex handwritten notes. To ensure a fair comparison with traditional OCR engines and to mitigate the risk of generative hallucinations (as observed with LightOnOCR), the VLM was guided by a strict zero-shot system prompt. The model was explicitly instructed to act solely as a medical OCR system, to preserve original line breaks, to strictly avoid any summarization or interpretation, and to output an [UNK] token for completely illegible words.

The aggregated metrics resulting from this strict extraction prompting are summarized in Table 3.4.

Table 3.4.: Task 1 pure transcription performance of the end-to-end VLM (Qwen3-VL-8B, 4-bit) on the stratified evaluation subset ($N = 48$). All metrics are expressed in percentages (%).

Metric	Mean	Median	Std. Dev.	Range
Accuracy	87.74	99.10	21.51	4.67 – 100.00
CER	3.81	0.75	7.45	0.00 – 47.60
WER	11.90	5.05	14.87	0.00 – 90.77
NED	3.74	0.75	7.11	0.00 – 44.90
Vocab F1-Score	88.71	93.30	13.01	23.21 – 100.00

The results reveal a paradigm shift in text recognition performance. Qwen3-VL-8B (4-bit) achieves a mean exact sequence accuracy of 87.74 %, demonstrating an extraordinary capacity to read highly unstructured handwriting. The Character Error Rate (CER) drops to an unprecedented mean of 3.81 %, while the Word Error Rate (WER) stabilizes at 11.90 %. Unlike traditional OCR systems, the VLM appears highly robust against background noise and layout variations. Furthermore, the model maintains a strong Vocabulary F1-Score of 88.71 %, providing an optimal foundation for the direct, zero-shot information extraction evaluated in Task 2.

Figure 3.4 visually confirms this high perceptual accuracy, demonstrating that for the vast majority of the test set, the character-level errors remain negligible.

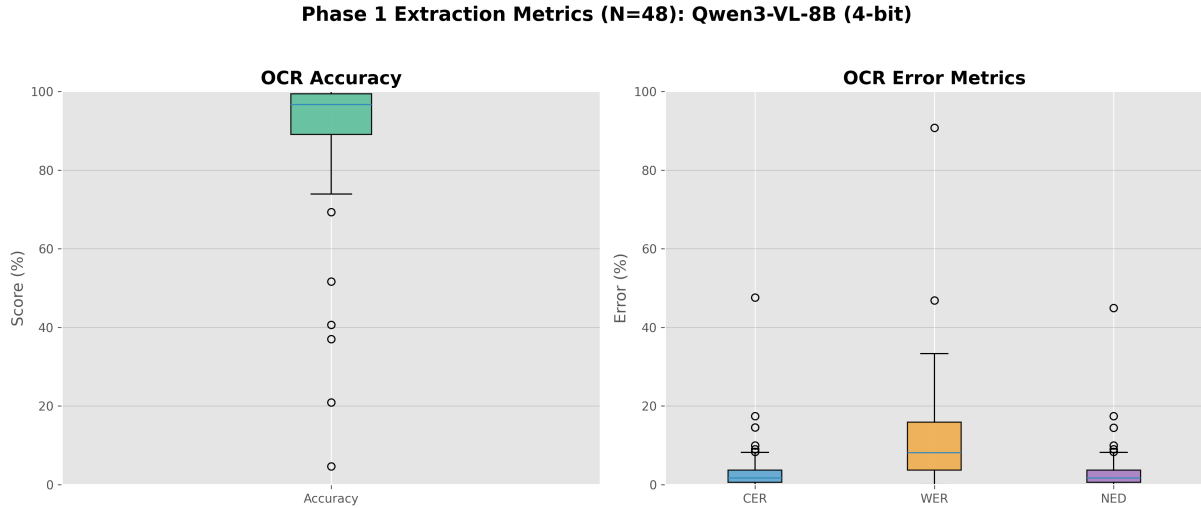


Figure 3.4.: Distribution of Task 1 transcription metrics for Qwen3-VL-8B (4-bit) across the $N = 48$ evaluation set. The compact distribution of the Character Error Rate (CER) near the zero baseline highlights the model’s high perceptual accuracy, despite occasional performance drops on highly ambiguous edge cases.

3.2.2.3. Evaluation of GPT-4o-mini

Evaluated under the identical zero-shot strict prompting conditions, the proprietary GPT-4o-mini model demonstrates a notably different behavior profile. The results are summarized in Table 3.5.

While still outperforming most traditional OCR engines, GPT-4o-mini records higher error rates than the open-weight Qwen model, with a mean CER of 11.84% and a substantially higher WER of 25.19%. This discrepancy suggests that while proprietary API models excel at general multimodal reasoning, their internal guardrails and generative priors often cause them to struggle when forced to adhere to strict, character-by-character reconstruction tasks in degraded visual contexts.

Table 3.5.: Task 1 pure transcription performance of GPT-4o-mini on the stratified evaluation subset ($N = 48$). All metrics are expressed in percentages (%).

Metric	Mean	Median	Std. Dev.	Range
Accuracy	69.24	85.35	32.64	4.65 – 100.00
CER	11.84	4.50	17.76	0.00 – 73.32
WER	25.19	16.20	26.16	0.00 – 103.08
NED	11.84	4.50	17.76	0.00 – 73.32
Vocab F1-Score	76.40	81.50	24.03	6.06 – 100.00

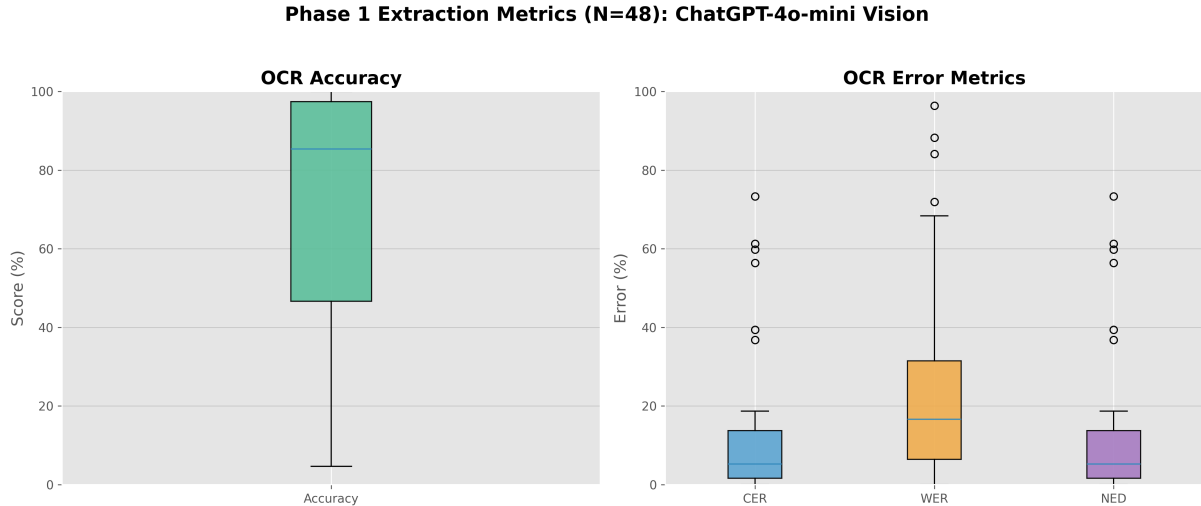


Figure 3.5.: Distribution of Task 1 transcription metrics for GPT-4o-mini across the $N = 48$ evaluation set. The CER distribution shows a right-skewed pattern with median 5.25% and several high-error outliers, indicating robust performance on most samples but challenges with complex edge cases.

3.2.2.4. Evaluation of Pixtral-12B

As the third end-to-end VLM candidate, the open-weight Pixtral-12B model was evaluated under the identical zero-shot strict prompting conditions. The aggregated results are presented in Table 3.6.

Table 3.6.: Task 1 pure transcription performance of Pixtral-12B on the stratified evaluation subset ($N = 48$). All metrics are expressed in percentages (%).

Metric	Mean	Median	Std. Dev.	Range
Accuracy	79.74	92.40	25.16	9.28 – 100.00
CER	9.18	3.60	16.27	0.00 – 70.43
WER	20.41	14.65	23.26	0.00 – 100.00
NED	8.67	3.55	14.62	0.00 – 61.28
Vocab F1-Score	83.13	88.45	19.10	23.53 – 100.00

Pixtral-12B achieves a mean exact sequence accuracy of 79.74%, placing it between the two previously evaluated models. Its mean CER of 9.18% and mean WER of 20.41% indicate a solid transcription capability that surpasses GPT-4o-mini but falls short of the exceptional performance demonstrated by Qwen3-VL-8B (4-bit). The model also achieves a strong Vocabulary F1-Score of 83.13%, demonstrating good semantic retention across the evaluation set.

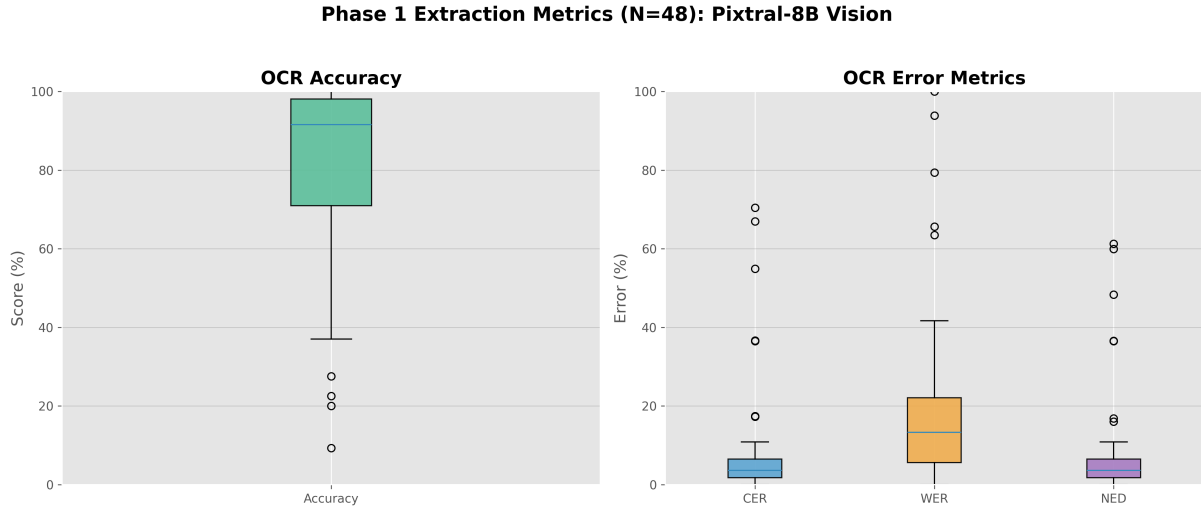


Figure 3.6.: Distribution of Task 1 transcription metrics for Pixtral-12B across the $N = 48$ evaluation set. Pixtral demonstrates superior median performance with the tightest CER/WER distributions near zero, confirming its leading position among VLMs for document text extraction.

3.2.3. Comparison of All Text Extractors and Transition to Task 2

EasyOCR suffers from severe baseline noise and struggles with fundamental character recognition, resulting in consistently high error rates across the board. Surya OCR provides the most structurally constrained and balanced output; it drastically improves character recognition but occasionally suffers from insertion errors. LightOnOCR-2, representing a specialized VLM-based OCR architecture, delivers state-of-the-art median accuracy and strong vocabulary retention, but is prone to explosive hallucinations on difficult edge cases.

Table 3.7 summarizes the performance of the three classical text extractors evaluated on the full stratified test set ($N = 559$). The comparison reveals that each architecture exhibits a highly distinct error profile.

Table 3.7.: OCR comparison on the full stratified test set ($N = 559$ valid samples). Performance is compared across traditional (EasyOCR), modern deep learning (Surya OCR), and specialized VLM-based (LightOnOCR-2) architectures. All values are given as mean and median percentages (%).

Metric	EasyOCR		Surya OCR		LightOnOCR-2	
	Mean	Median	Mean	Median	Mean	Median
Accuracy	13.62	9.07	62.96	72.58	65.62	82.43
CER	60.67	65.49	14.43	5.05	40.21	4.44
WER	94.45	98.15	43.72	27.27	67.14	21.31
NED	60.66	65.49	13.15	5.04	17.16	4.44
Vocab F1-Score	12.92	6.11	65.85	75.11	70.10	80.00

The comparison reveals a consistent performance ranking across all five metrics. The open-weight Qwen3-VL-8B (4-bit) achieves the best transcription fidelity, attaining the lowest mean CER (3.81 %) and the highest Vocabulary F1-Score (88.71 %). Pixtral-12B ranks second, de-

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livering competitive performance with a mean CER of 9.18 %, while GPT-4o-mini records the weakest pure OCR scores with a mean CER of 11.84 % and a WER of 25.19 %. Notably, all three VLMs substantially outperform the classical EasyOCR baseline (mean CER: 60.67 %) and demonstrate performance comparable to or exceeding Surya OCR (mean CER: 14.43 %), confirming the capability of modern Vision-Language architectures as competitive alternatives to dedicated OCR pipelines.

Table 3.8.: VLM Transcription Performance Comparison (N=48 stratified samples). Metrics shown as Mean and Median percentages (%).

Metric	Qwen3-VL-8B (4-bit)		Pixtral-12B		GPT-4o-mini	
	Mean	Median	Mean	Median	Mean	Median
Accuracy	87.75	91.60	79.74	96.60	69.24	85.35
CER	3.81	3.60	9.18	1.65	11.85	5.25
WER	11.89	13.35	20.41	8.10	25.18	16.60
NED	3.75	3.60	8.67	1.65	11.85	5.25

Taken together, these findings motivate two complementary downstream evaluation strategies in Task 2. First, the diverse error profiles of the three classical OCR models (high noise, structural insertions, and specialized VLM hallucinations) offer a unique opportunity to test the robustness and error-correction capabilities of textual Large Language Models (Approach 1: OCR + LLM pipelines). Second, the superior transcription performance of the end-to-end VLMs justifies a direct evaluation of their zero-shot summarization and information extraction capabilities from raw images, entirely bypassing the classical OCR step (Approach 2: end-to-end Vision-Language Models).

3.3. Task 2: Downstream Understanding and Extraction

Building upon the distinct error profiles of the text extractors established in Task 1, this section evaluates the downstream understanding and information extraction capabilities of the overall pipelines. The primary objective is to investigate how effectively Large Language Models (LLMs) can parse, reconstruct, and extract structured medical data from noisy or hallucinated OCR transcriptions.

”As defined and detailed in Section 3.2.2.1, the Task 2 downstream evaluation of the end-to-end VLMs and OCR+LLM pipelines was conducted on the exact same $N = 48$ stratified subset to ensure a fair comparison.”

3.3.1. Approach 1: OCR + LLM Pipelines

The primary objective of Approach 1 is to investigate how effectively Large Language Models (LLMs) can parse, reconstruct, and extract structured medical data from the noisy, hallucinated, or fragmented OCR transcriptions generated in Task 1.

To systematically analyze the error-correction capabilities of the reasoning models, the outputs of the three text extractors—EasyOCR (high base noise), Surya OCR (structural insertions), and LightOnOCR-2 (generative hallucinations)—were fed directly into three distinct LLMs in a zero-shot setting:

3.3.1.1. Results on High-Noise Input (EasyOCR)

The first phase of the downstream evaluation analyzes the pipelines based on the EasyOCR baseline, which represents a scenario of extreme noise and fragmentation (mean WER: 94.45 %). The objective here is to evaluate the extent to which the LLMs can grammatically and semantically reconstruct highly degraded text and still extract the relevant medical concepts. The aggregate results across the 48 evaluated samples are summarized in Table 3.9.

The general text reconstruction metrics reveal that massive character-level degradation leads to low performance across all three language models. ROUGE scores remain exceptionally low; the highest ROUGE-1 F1 score is achieved by GPT-4o-mini at only 19.89 %. Similarly, the BERTScore F1 metric drops to 8.17 % for Ministral-8B and peaks at 17.16 % for GPT-4o-mini. The missing syntactic structure in the EasyOCR output prevents the LLMs from reliably reconstructing the original narrative flow.

Table 3.9.: Pipeline evaluation results using EasyOCR as the text extractor ($N = 48$). Performance is compared across Qwen3-8B (4-bit), Ministral-8B, and GPT-4o-mini in a zero-shot setting. All metrics are given as mean percentages (%).

Metric	Qwen3-8B (4-bit)	Ministral-8B	GPT-4o-mini
ROUGE-1 F1	17.09	14.53	19.89
ROUGE-2 F1	4.83	4.70	6.16
ROUGE-L F1	11.94	10.16	14.65
BERTScore Precision	10.95	2.74	18.46
BERTScore Recall	14.45	13.81	15.81
BERTScore F1	12.74	8.17	17.16
Med.Term Precision	37.86	38.91	42.34
Med.Term Recall	52.57	59.47	51.83
Med.Term F1	41.19	43.42	44.48

A highly contrasting pattern emerges when analyzing the targeted extraction of medical terminology. Despite the highly fragmented inputs, all models achieve a Medical Term F1-Score above 41 %. The compact Ministral-8B model achieves the highest Medical Term Recall (59.47 %), indicating a strong capability to recognize and extract potential entities from garbled text sequences. GPT-4o-mini provides the most balanced extraction capability, achieving the highest Med.Term Precision (42.34 %) and the best overall Med.Term F1-Score (44.48 %). Qwen3-8B (4-bit) performs competitively, settling between the other two models with a Med.Term F1-Score of 41.19 %.

Overall, the EasyOCR-based pipelines show that while severe OCR noise strongly impedes global text reconstruction, the evaluated LLMs maintain a moderate capacity to isolate and recover core clinical entities.

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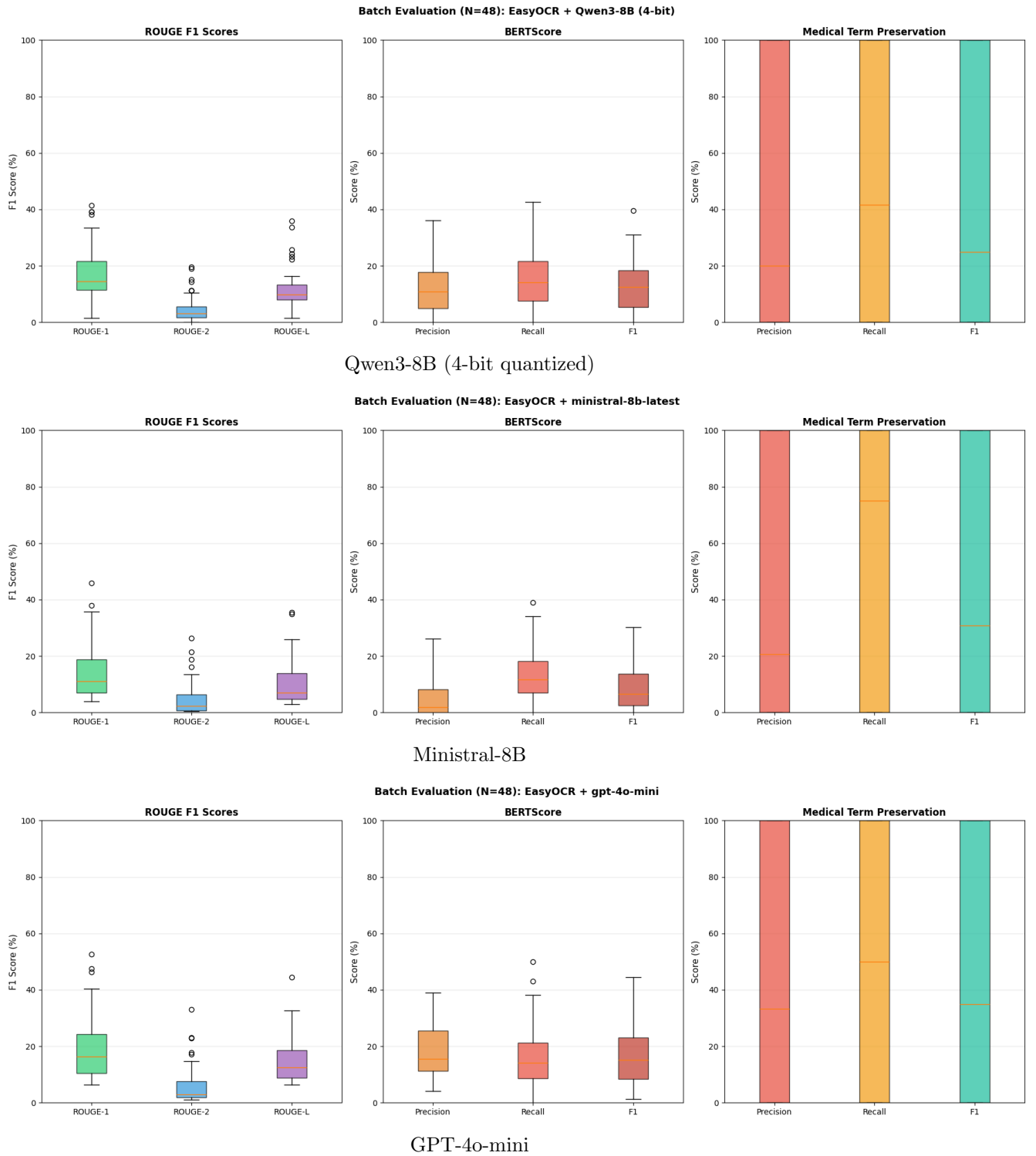


Figure 3.7.: Distribution of ROUGE F1, BERTScore, and Medical Term Preservation across the 48 evaluation samples for the EasyOCR pipelines.

3.3.1.2. Results on Structurally Constrained Input (Surya OCR)

The second phase of the downstream evaluation utilizes the transcriptions from Surya OCR. As established in Task 1, this text extractor provides a much stronger structural and character-level foundation (mean CER: 14.43 %), although it occasionally suffers from over-segmentation and insertion errors. The goal is to evaluate how the LLMs handle these structurally robust but partially inflated inputs. The aggregate results across the 48 samples are presented in Table 3.10.

Table 3.10.: Pipeline evaluation results using Surya OCR as the text extractor ($N = 48$). Performance is compared across Qwen3-8B (4-bit), Ministral-8B, and GPT-4o-mini in a zero-shot setting. All metrics are given as mean percentages (%).

Metric	Qwen3-8B (4-bit)	Ministral-8B	GPT-4o-mini
ROUGE-1 F1	35.93	34.08	39.54
ROUGE-2 F1	17.97	17.15	19.38
ROUGE-L F1	27.01	27.08	30.26
BERTScore Precision	22.88	15.30	27.32
BERTScore Recall	34.57	33.65	37.19
BERTScore F1	28.59	24.08	32.16
Med.Term Precision	50.38	55.31	53.87
Med.Term Recall	68.83	79.37	74.32
Med.Term F1	54.99	61.06	59.49

Compared to the EasyOCR baseline, the improved input quality provided by Surya OCR leads to a substantial increase across all performance metrics for all three models. The general text reconstruction metrics show a distinct upward shift. ROUGE-1 F1 scores rise to approximately 34 % for Ministral-8B and nearly 36 % for Qwen3-8B (4-bit), while GPT-4o-mini achieves the highest score at 39.54 %. The semantic reconstruction, measured by BERTScore F1, follows a similar trend, improving significantly compared to the high-noise scenario. GPT-4o-mini leads this category with a BERTScore F1 of 32.16 %, followed by Qwen3-8B (4-bit) at 28.59 % and Ministral-8B at 24.08 %.

The targeted extraction of medical terminology also benefits greatly from the more robust textual foundation. All three models achieve a Medical Term F1-Score above 54 %. Consistent with the previous findings, the compact Ministral-8B model exhibits a highly aggressive extraction behavior, yielding the highest Medical Term Recall (79.37 %) and consequently the highest overall Medical Term F1-Score (61.06 %). GPT-4o-mini maintains a balanced extraction profile with an F1-Score of 59.49 %, while Qwen3-8B (4-bit) reaches an F1-Score of 54.99 %.

The results demonstrate that when provided with structurally intact text—even if it contains insertion errors—the LLMs are highly capable of filtering the relevant information and reconstructing both the general narrative and the specific clinical entities.

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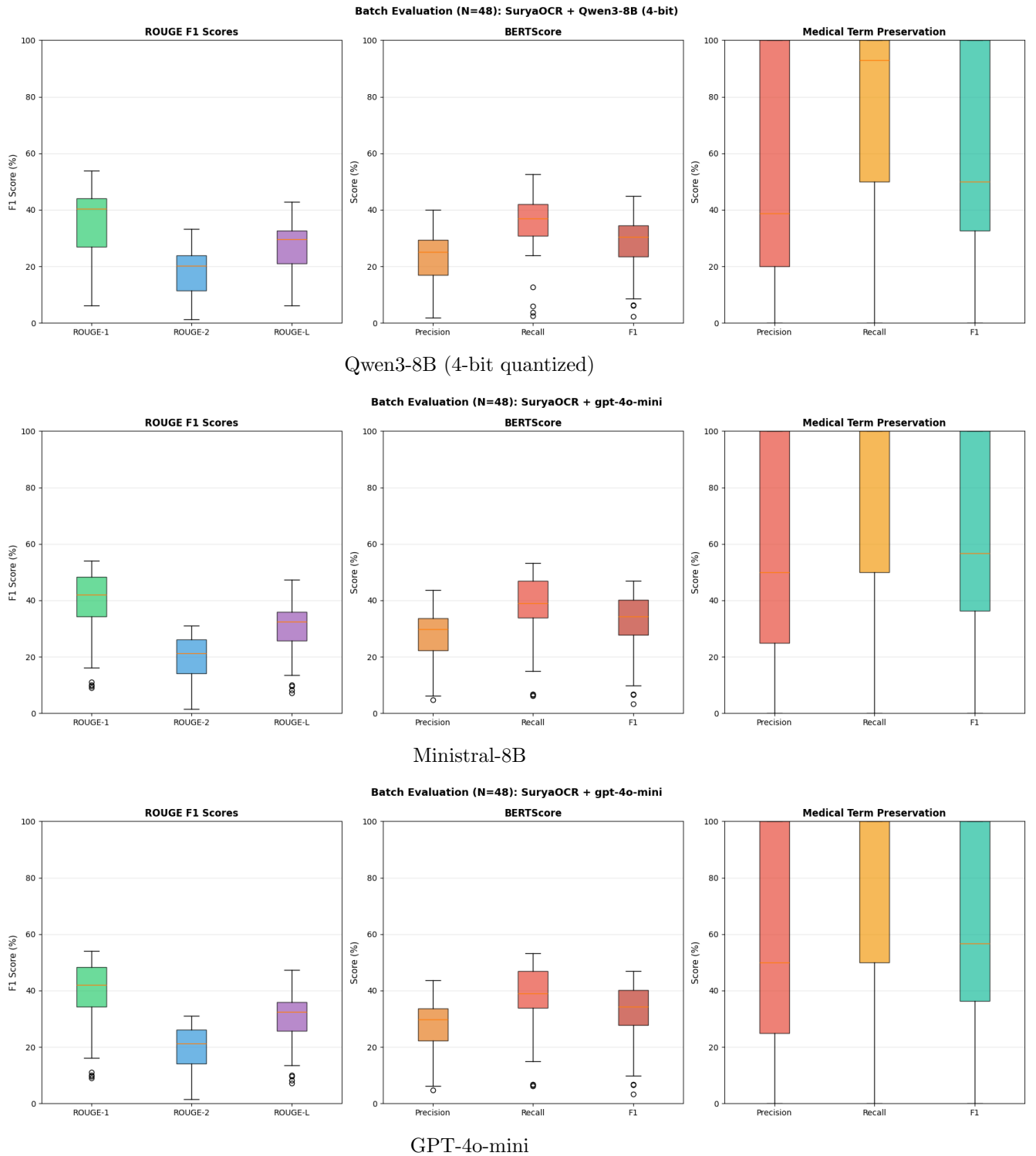


Figure 3.8.: Distribution of ROUGE F1, BERTScore, and Medical Term Preservation across the 48 evaluation samples for the Surya pipelines.

3.3.1.3. Results on Generative VLM Input (LightOnOCR-2)

The final phase of Approach 1 evaluates the LLM pipelines using the output from LightOnOCR-2. As demonstrated in Task 1, this Vision-Language Model provides the highest median exact sequence accuracy but is prone to generating extensive hallucinations when encountering illegible text blocks. The objective of this evaluation is to determine how downstream LLMs cope with highly accurate but potentially hallucinated text fragments. The performance across the 48 samples is summarized in Table 3.11.

Table 3.11.: Pipeline evaluation results using LightOnOCR as the text extractor ($N = 48$). Performance is compared across Qwen3-8B (4-bit), Ministral-8B, and GPT-4o-mini in a zero-shot setting. All metrics are given as mean percentages (%).

Metric	Qwen3-8B (4-bit)	Ministral-8B	GPT-4o-mini
ROUGE-1 F1	37.79	34.58	41.56
ROUGE-2 F1	18.45	17.34	21.11
ROUGE-L F1	28.83	27.74	31.01
BERTScore Precision	24.39	16.10	29.44
BERTScore Recall	35.96	33.47	39.83
BERTScore F1	30.07	24.44	34.54
Med.Term Precision	53.02	55.34	54.78
Med.Term Recall	73.63	78.85	78.63
Med.Term F1	58.51	61.70	61.67

The aggregated data reveals that the LightOnOCR transcriptions lead to the highest overall downstream performance among all evaluated OCR baselines. Both lexical and semantic reconstruction metrics reach their peak values in this setup. GPT-4o-mini achieves the highest ROUGE-1 F1 score (41.56 %) and BERTScore F1 (34.54 %). Qwen3-8B (4-bit) follows closely with a ROUGE-1 F1 of 37.79 % and a BERTScore F1 of 30.07 %, outperforming Ministral-8B in terms of general text reconstruction.

Regarding the targeted extraction of medical terminology, the models demonstrate an exceptional ability to filter the correct entities from the VLM-generated text. The Medical Term F1-Scores for both Ministral-8B (61.70 %) and GPT-4o-mini (61.67 %) are nearly identical, representing the highest entity preservation rates recorded in the OCR+LLM evaluation. Ministral-8B again records the highest Medical Term Recall (78.85 %), but GPT-4o-mini is positioned only marginally behind (78.63 %) while maintaining a slightly higher precision. Qwen3-8B (4-bit) also delivers robust results with a Medical Term F1-Score of 58.51 %.

These results indicate that state-of-the-art LLMs possess a high tolerance for the generative hallucinations produced by LightOnOCR-2. Instead of being derailed by hallucinated text blocks, the reasoning models effectively leverage the underlying high-accuracy character recognition of the VLM to isolate and extract the relevant clinical information.

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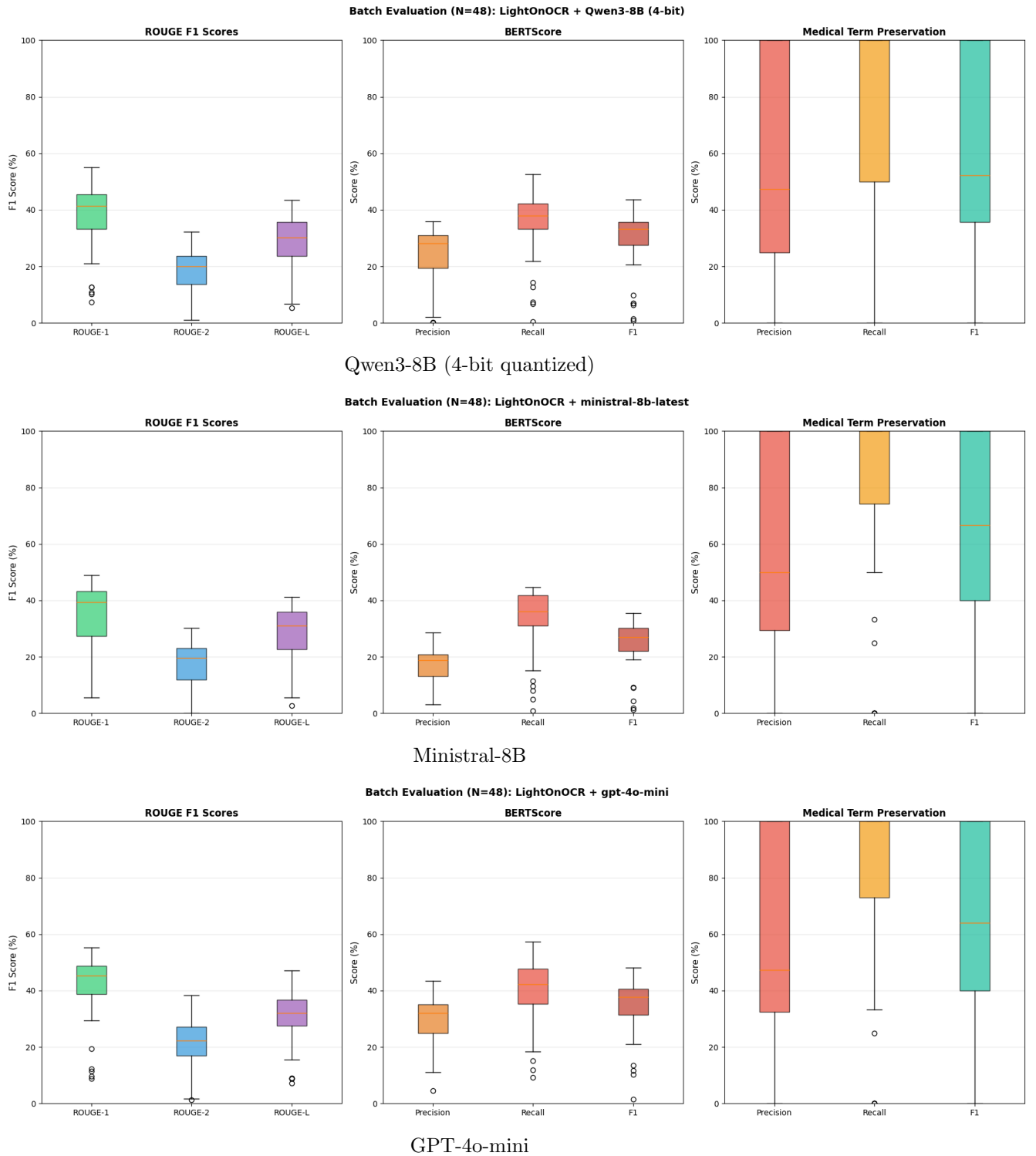


Figure 3.9.: Distribution of ROUGE F1, BERTScore, and Medical Term Preservation across the 48 evaluation samples for the LightOnOCR-2 pipelines.

3.3.1.4. Summary of OCR + LLM Pipelines

The systematic evaluation of Approach 1 demonstrates a clear dependency between the quality of the primary text extraction and the downstream performance of the Language Models. When confronted with heavily degraded input (EasyOCR), the LLMs fail to reconstruct the semantic context, though they maintain a baseline ability to salvage isolated medical entities. Providing structurally constrained input (Surya OCR) leads to a significant performance leap across all metrics. Surprisingly, the highest downstream scores are achieved when reasoning models process the output of a Vision-Language Model (LightOnOCR). The LLMs prove highly resilient to VLM-induced hallucinations, effectively leveraging the high character-level accuracy to extract clinical entities with an F1-Score exceeding 61 %.

While this two-step architecture (OCR + LLM) yields robust extraction results, it introduces inherent structural bottlenecks: OCR errors cascade directly into the LLM prompt, and spatial layout information is completely lost during the text serialization phase. These limitations motivate the investigation of Approach 2 in the following section, where end-to-end Vision-Language Models are deployed to process the raw handwritten images directly, bypassing the traditional OCR intermediate step.

3.3.2. Approach 2: end-to-end Vision-Language Models (VLMs)

In this section, we evaluate the performance of end-to-end Vision-Language Models (VLMs) on Task 2, focusing on their ability to accurately extract and structure clinical information. The evaluation is based on lexical similarity (ROUGE), semantic equivalence (BERTScore), and domain-specific accuracy (Medical Term extraction).

3.3.2.1. Qwen3-VL-8B (4-bit)

The Qwen3-VL model demonstrates strong capabilities in general text generation and information retrieval. As shown in Table 3.12, it achieves a mean ROUGE-1 F1-score of 41.15% and a BERTScore F1 of 35.92%. Its most notable strength lies in identifying domain-specific terminology, recording a high Medical Term Recall of 80.85%. However, its Medical Term Precision is noticeably lower at 57.28%, suggesting a tendency to over-extract. The high standard deviation across metrics, alongside minimum scores of 0.0% and maximums of 100.0% in medical term extraction, reveals significant performance fluctuations depending on the individual sample.

Table 3.12.: Task 2 information extraction performance for **Qwen3-VL-8B (4-bit)** ($N = 48$).

Metric	Mean (%)	Std (%)	Min (%)	Max (%)
ROUGE-1 F1	41.15	9.02	17.12	59.26
ROUGE-2 F1	20.47	7.50	4.55	40.28
ROUGE-L F1	31.86	9.50	11.71	53.42
BERTScore Precision	30.28	7.80	10.13	44.85
BERTScore Recall	41.79	7.73	15.84	55.01
BERTScore F1	35.92	7.26	13.07	48.34
Med.Term Precision	57.28	32.63	0.00	100.00
Med.Term Recall	80.85	29.34	0.00	100.00
Med.Term F1	64.14	29.84	0.00	100.00

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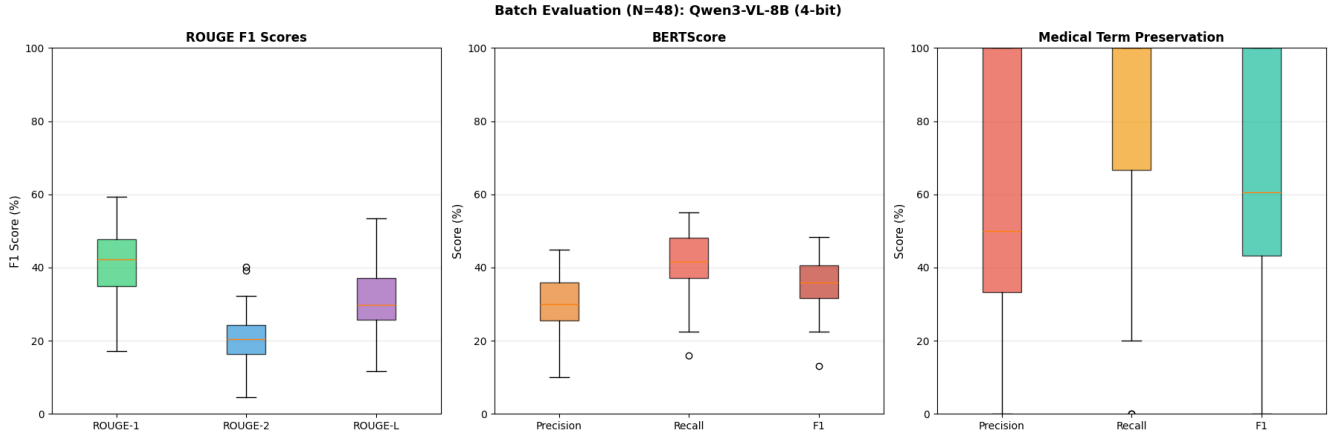


Figure 3.10.: Distribution of Task 2 information extraction metrics for **Qwen3-VL-8B (4-bit)** across the $N = 48$ evaluation set. The model exhibits the highest recall for medical entities, though the wide distribution from 0% to 100% in the medical term metrics emphasizes the sample-dependent nature of the extraction.

3.3.2.2. Pixtral-12B

Pixtral-12B shows a balanced performance profile (Table 3.13). It achieves the highest ROUGE-2 F1-score of 20.62% among the evaluated models, indicating good preservation of bigram phrasing. For clinical entity extraction, Pixtral achieves a Medical Term Precision of 58.94% and a Recall of 73.35%. Similar to Qwen3-VL, the standard deviation for the Medical Term F1 is exceptionally high at 34.58%, underscoring an inconsistency in handling complex or ambiguous document layouts.

Table 3.13.: Task 2 information extraction performance for **Pixtral-12B** ($N = 48$).

Metric	Mean (%)	Std (%)	Min (%)	Max (%)
ROUGE-1 F1	39.56	11.81	8.49	60.77
ROUGE-2 F1	20.62	8.91	0.95	39.11
ROUGE-L F1	31.36	9.28	7.55	49.72
BERTScore Precision	24.87	9.32	1.17	40.00
BERTScore Recall	39.35	10.41	8.51	58.34
BERTScore F1	31.91	9.53	4.90	48.84
Med.Term Precision	58.94	36.54	0.00	100.00
Med.Term Recall	73.35	35.44	0.00	100.00
Med.Term F1	62.47	34.58	0.00	100.00

3.3. Task 2: Downstream Understanding and Extraction

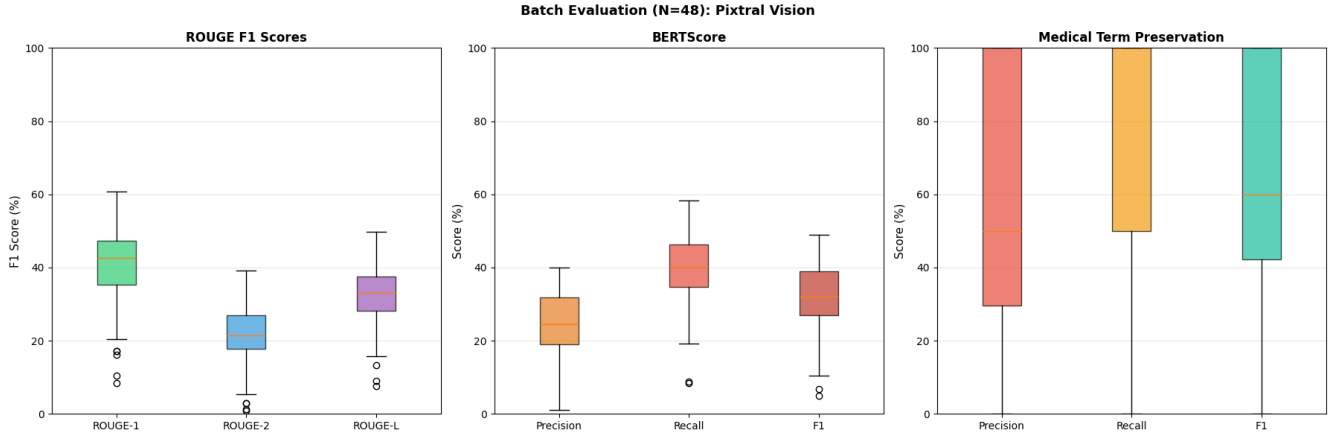


Figure 3.11.: Distribution of Task 2 information extraction metrics for **Pixtral Vision** across the $N = 48$ evaluation set. The plot illustrates a balanced profile between precision and recall. However, the large interquartile ranges for the domain-specific medical terms highlight the model’s inconsistency across varying document layouts.

3.3.2.3. GPT-4o-mini

As detailed in Table 3.14, GPT-4o-mini delivers competitive semantic equivalence metrics, achieving a BERTScore Precision of 33.79% and an overall BERTScore F1 of 35.15%. In domain-specific extraction, GPT-4o-mini exhibits a highly conservative extraction behavior: it achieves the highest Medical Term Precision (59.58%) among all evaluated models, indicating a low rate of false positives. However, this comes at the expense of recall, where it yields the lowest score of 69.96%, leading to an overall Medical Term F1-score of 61.49%. The variance remains high across all medical term metrics (e.g., a standard deviation of 33.30% for the F1-score), reiterating the layout-dependent nature of the extraction task.

Table 3.14.: Task 2 information extraction performance for **GPT-4o-mini** ($N = 48$).

Metric	Mean (%)	Std (%)	Min (%)	Max (%)
ROUGE-1 F1	38.77	11.20	8.79	54.55
ROUGE-2 F1	19.07	8.10	1.71	34.48
ROUGE-L F1	31.11	9.36	8.11	46.59
BERTScore Precision	33.79	7.80	12.83	46.69
BERTScore Recall	36.47	11.10	5.73	56.25
BERTScore F1	35.15	8.83	9.34	51.44
Med.Term Precision	59.58	35.43	0.00	100.00
Med.Term Recall	69.96	34.29	0.00	100.00
Med.Term F1	61.49	33.30	0.00	100.00

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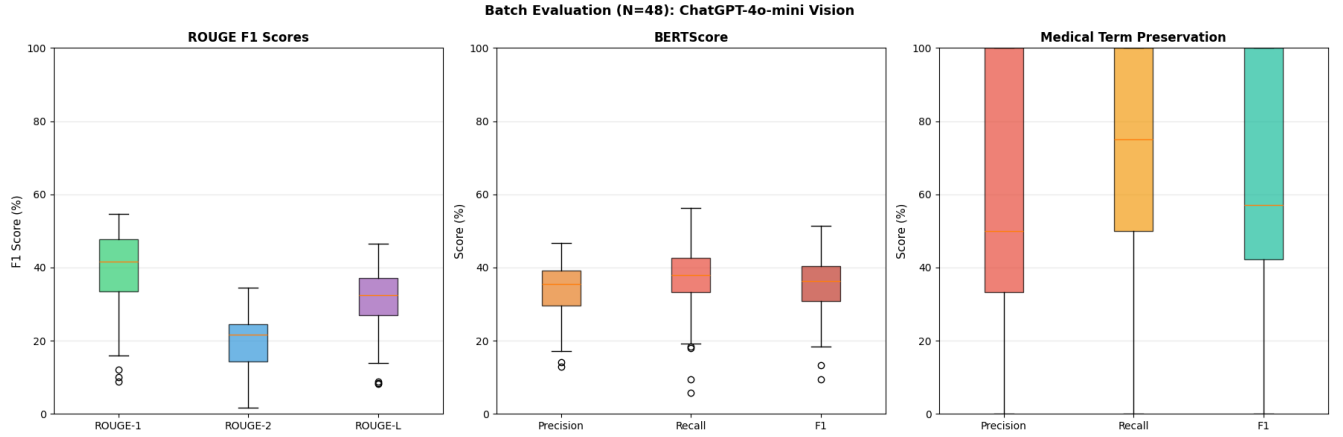


Figure 3.12.: Distribution of Task 2 information extraction metrics for **ChatGPT-4o-mini Vision** across the $N = 48$ evaluation set. The results demonstrate the model’s highly conservative extraction behavior, characterized by high precision but lower recall, alongside significant variance in correctly identifying medical entities.

3.3.2.4. Comparison of VLM Performances

Table 3.15 summarizes the aggregate mean and standard deviation of all evaluated VLMs for a direct side-by-side comparison. Overall, the three models perform at a remarkably similar level, though with distinct operational characteristics. Qwen3-VL emerges as the frontrunner in both general textual overlap (ROUGE metrics) and the Medical Term F1-score (64.14%), driven by its significantly superior recall (80.85%).

In contrast, GPT-4o-mini operates more conservatively. It provides the highest Medical Term Precision (59.58%) and BERTScore Precision (33.79%), making it the most reliable model when it comes to avoiding false positives, though it struggles with strict medical entity recall. Pixtral-12B positions itself as a balanced middle ground, maintaining competitive precision (58.94%) and recall (73.35%) while leading in ROUGE-2 phrasing (20.62%).

Table 3.15.: Task 2 information extraction performance comparison across end-to-end VLMs ($N = 48$). Metrics are reported as Mean percentages (%).

Metric	Qwen3-VL-8B (4-bit)	Pixtral-12B	GPT-4o-mini
ROUGE-1 F1	41.15	39.56	38.77
ROUGE-2 F1	20.47	20.62	19.07
ROUGE-L F1	31.86	31.36	31.11
BERTScore Precision	30.28	24.87	33.79
BERTScore Recall	41.79	39.35	36.47
BERTScore F1	35.92	31.91	35.15
Med.Term Precision	57.28	58.94	59.58
Med.Term Recall	80.85	73.35	69.96
Med.Term F1	64.14	62.47	61.49

A critical takeaway across all three VLMs is the immense variance in performance (reflected in standard deviations consistently exceeding 30% for Medical Terms). This highlights that while

VLMs can perfectly extract clinical data from certain documents, they lack the consistency required for fully automated, unverified medical structuring.

3.3.2.5. Impact of Few-Shot Prompting

While the primary evaluation of the VLMs was conducted in a zero-shot setting to establish a baseline, supplementary experiments investigated the effect of in-context learning. Providing the models with one to two few-shot examples of correctly extracted clinical JSON structures resulted in a marginal but consistent performance increase of approximately 1% to 2% in the overall Medical Term F1-scores across all models.

3.3.3. Overall Comparison: Pipeline vs. End-to-End VLMs

Evaluating the extraction of clinical information (Task 2) revealed a clear performance shift between the decoupled OCR-LLM pipeline (Approach 1) and the end-to-end Vision-Language Models (Approach 2). A direct comparison between the best-performing pipeline configuration (LightOnOCR combined with various LLMs) and the standalone VLMs highlights the superiority of the end-to-end multimodal approach.

The most critical metric for this task, the Medical Term F1-score, distinctly favors the VLMs. While the best pipeline setups (Minstral-8B and GPT-4o-mini) plateaued at Medical Term F1-scores of 61.70% and 61.67% respectively, the end-to-end VLMs surpassed this threshold. Qwen3-VL-8B achieved a leading score of 64.14%, followed closely by Pixtral-12B at 62.47%. This improvement is primarily driven by the VLMs' superior Medical Term Precision (peaking at 59.58% for GPT-4o-mini in VLM mode, compared to 55.34% for the pipeline), indicating that the integrated visual-semantic processing effectively reduces false-positive extractions.

Furthermore, the end-to-end architecture eliminates compounding errors inherent to the two-step pipeline. In Approach 1, any text missed or hallucinated by the OCR engine (LightOnOCR) cannot be recovered by the subsequent LLM, inherently capping the system's recall. VLMs bypass this bottleneck by grounding their extraction directly in the original visual input, allowing them to contextually infer ambiguous or poorly scanned terms that a pure OCR engine might drop.

Conclusion for Deployment: Based on its superior performance in domain-specific entity extraction and the architectural advantage of eliminating intermediate OCR bottlenecks, the end-to-end VLM approach proves to be the more effective solution. While Qwen3-VL-8B (4-bit) demonstrated the highest overall metrics, **Pixtral-12B** was ultimately selected as the primary extraction engine for real-world integration onto the robotic platform.

This decision is driven by two critical practical constraints:

- **Hardware Limitations:** The computational resources available directly on the mobile robotic platform are insufficient to run large multimodal models locally at an acceptable inference speed. Consequently, the extraction process must be offloaded to a cloud-based API.
- **Data Privacy and Deployment Complexity:** Processing highly sensitive clinical documents demands strict adherence to European data protection regulations (GDPR). Relying on global third-party aggregators (e.g., OpenRouter) to access models like Qwen3-VL introduces severe privacy risks, as it obscures the physical processing location of patient data. While deploying Qwen3-VL on dedicated European GPU servers or utilizing recently

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established EU-based Alibaba Cloud endpoints is technically feasible to ensure GDPR compliance, these approaches introduce significant infrastructural overhead and maintenance complexity.

In contrast, Pixtral-12B is natively provided by a European AI company (Mistral AI) with official, GDPR-compliant enterprise APIs (La Plateforme) hosted exclusively on European servers. This "out-of-the-box" data sovereignty eliminates the need for complex self-hosting infrastructure while guaranteeing legal compliance, making it the most pragmatic and secure cloud-based solution for this use case compared to US-based (GPT-4o) or Chinese-developed (Qwen) models.

By prioritizing Pixtral-12B, the deployed system successfully balances state-of-the-art multi-modal extraction accuracy with the strict hardware and privacy requirements of real-world clinical robotics.

3.3.4. System Integration and Interactive Proof-of-Concept

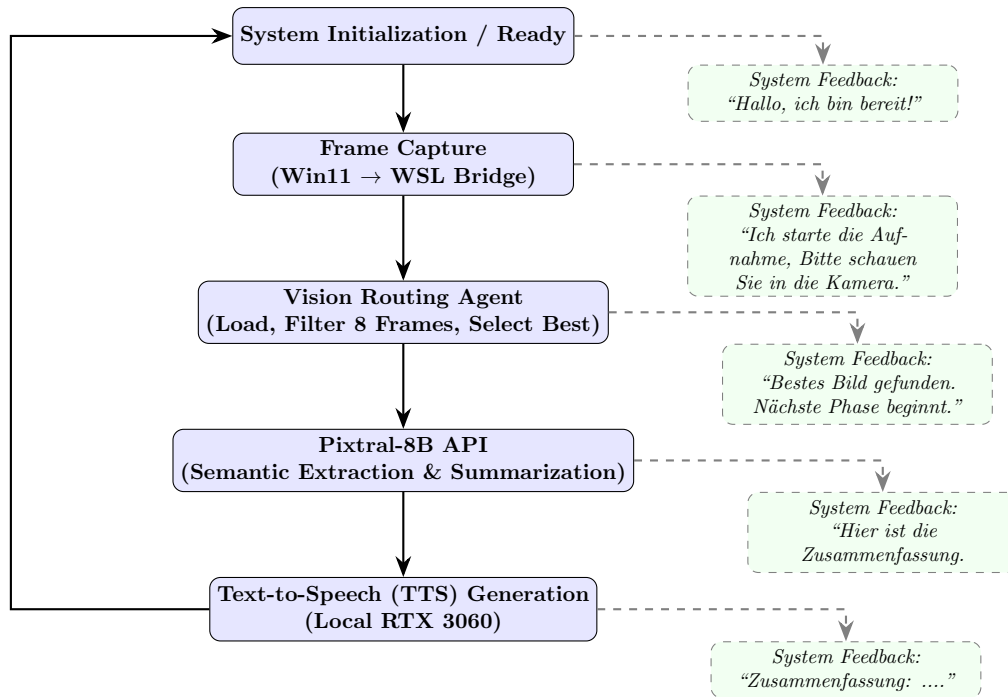


Figure 3.13.: Sequential workflow of the interactive proof-of-concept. The primary processing states (blue) are continuously accompanied by cached acoustic system feedback (green, dashed) to bridge hardware-induced latencies. Upon completion, the system loops back to the ready state after vocalizing the summary.

To validate the practical applicability of the proposed solution, the overall best-performing model (Pixtral-8B via API) was integrated into the Python-based proof-of-concept software. The demonstration was conducted on a local computing setup utilizing a Windows Subsystem for Linux (WSL) environment. Because WSL inherently lacks direct hardware access to USB peripherals like webcams, a dedicated Python bridge-server was implemented on the Windows 11 host. This server captures the hardware video stream and forwards the frames into the WSL environment, ensuring a seamless connection between the physical sensors and the Linux-based AI pipeline.

3.3.4.1. Inference Time and System Latency

A critical factor for the acceptance of a robotic assistant in clinical practice is its response time. During the initial system start, a complete end-to-end execution—from image capture to the spoken summary—takes approximately 3 minutes and 12 seconds.

This total duration is heavily dominated by the local Text-to-Speech (TTS) generation. Executed on a local NVIDIA RTX 3060 GPU, the process of loading the TTS model weights into VRAM and generating the lengthy audio file for the final clinical summary requires approximately 2 minutes and 22 seconds on its first run. The remaining time is distributed among the highly efficient analytical components: capturing a burst of 8 frames and running the local Vision Routing Agent for visual triage takes roughly 10 seconds, while the cloud-based Pixtral-8B API returns the structured JSON summary within 2 to 5 seconds depending on text density. For subsequent documents, once the model weights are cached in the VRAM, the overall processing time decreases significantly.

To ensure user acceptance despite the initial hardware-induced TTS bottleneck, continuous acoustic status updates were implemented to transparently bridge the processing gaps.

3.3.4.2. Robustness to Environmental Noise and Hardware

In contrast to the perfectly scanned documents used during the benchmark evaluation ($N = 48$), practical deployment involves capturing images via standard optical sensors. For this proof-of-concept, an LG AN-VC500 webcam was utilized.

The demonstration revealed that the multimodal architecture handles environmental artifacts effectively, provided a baseline of illumination is met. The Vision Routing Agent’s frame-selection algorithm proved highly dependent on ambient lighting; under poor lighting conditions, the image quality score frequently dropped below the predefined threshold between 30-60, triggering the *Human-in-the-Loop* protocol to request a better capture. However, under adequate illumination, the system demonstrated exceptional robustness. Tests with newly created, authentic handwritten clinical notes yielded a 100% successful extraction and summarization rate (an example extraction is illustrated in Figure 3.15). This confirms that the combination of local frame routing and cloud-based VLM extraction can reliably process unstandardized handwriting in real-world scenarios.

3.3.4.3. Human-Robot Interaction (HRI) Interface

A successful robotic deployment in a stressful clinical environment requires an intuitive and highly responsive user interface. To elevate the proof-of-concept from a mere data pipeline to an interactive assistant, a dedicated Human-Robot Interaction (HRI) layer was engineered, focusing on latency optimization and user engagement.

To establish a consistent persona for the robotic assistant, the Qwen-TTS-Base-1.7 model was utilized to create a custom cloned voice. This allows clinical facilities to personalize the assistant’s voice profile. To further optimize the previously discussed system latency, all recurring status updates (e.g., *“I am starting the recording”* or *“I have found the best image”*) are pre-generated and cached locally. This hybrid audio approach ensures instantaneous acoustic feedback for standard state transitions, while reserving the computationally expensive dynamic TTS generation exclusively for unpredictable `Ask_Human` queries (e.g., verifying specific medication names).

3. Experiments and Results

Furthermore, the acoustic feedback is complemented by a visual, context-aware graphical interface. As illustrated in Figure 3.14, the system features a minimalistic, animated digital face that changes expressions dynamically based on the current process state. For instance, the face displays a neutral, focused expression during document scanning, transitions to a thinking animation during API inference, and adopts an open, speaking expression when querying the nurse. This visual anthropomorphism provides immediate, non-verbal state feedback to the user, bridging waiting times and making the robotic interaction significantly more natural and approachable for clinical staff.

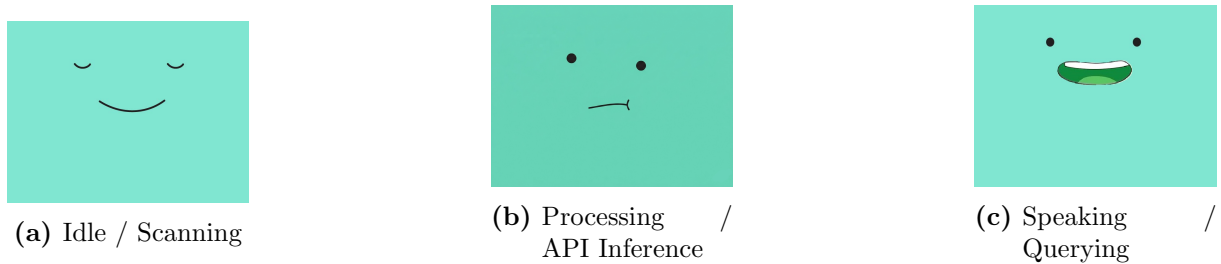
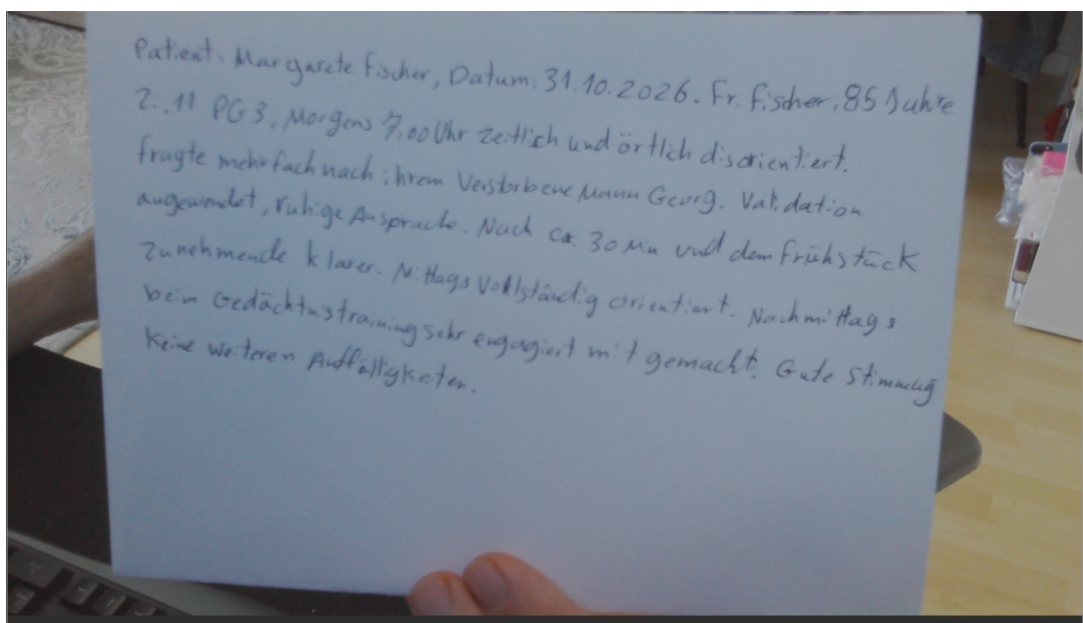
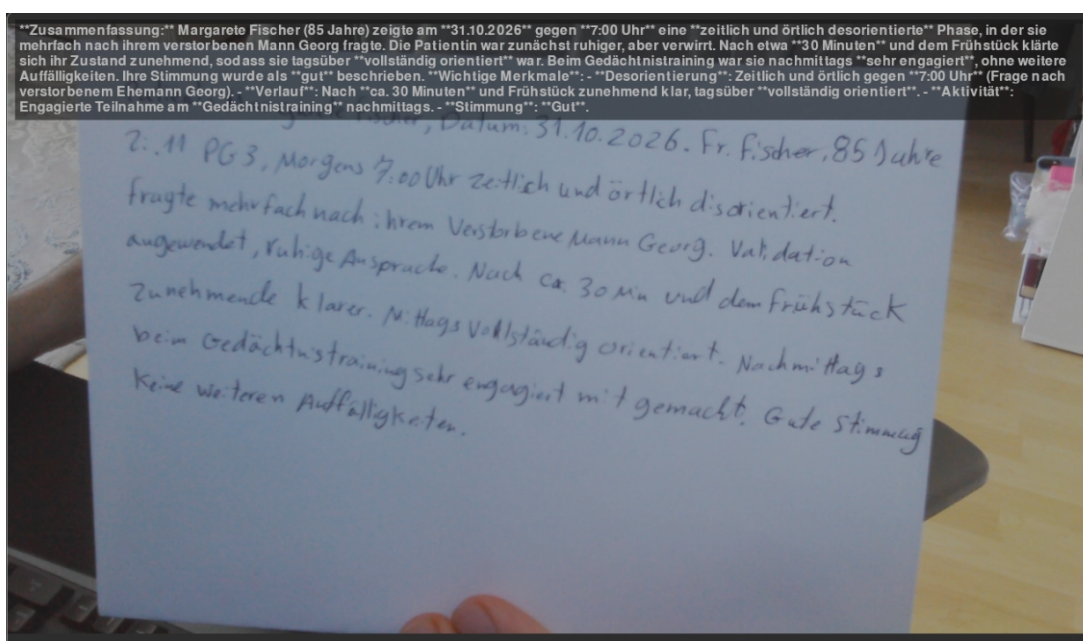


Figure 3.14.: The digital persona of the Human-Robot Interaction (HRI) interface. The animated face dynamically changes its expression to provide non-verbal, visual feedback corresponding to the system’s current processing state, thereby enhancing user trust and bridging inference latencies.



- (a) Raw input: A simulated clinical note captured via webcam (LG AN-VC500), displaying complex, unstandardized handwriting concerning a patient's cognitive state and daily activities.



- (b) System output: The structured and semantically accurate clinical summary generated by the Pixtral-8B API, successfully resolving chronological events, medication context, and patient mood without hallucinations.

Figure 3.15.: End-to-end execution of the interactive proof-of-concept. The Vision Routing Agent successfully selected the optimal frame from a burst capture, allowing the primary VLM to flawlessly extract and structure the handwritten text despite typical environmental noise and varying illumination.

4. Discussion

The objective of this study was to evaluate and deploy a robust information extraction system for clinical documents on a mobile robotic platform. By comparing a decoupled OCR-LLM pipeline against end-to-end Vision-Language Models (VLMs), this chapter interprets the results to explicitly answer the central research questions defined in the introduction regarding extraction accuracy, semantic comprehension, and systemic reliability.

4.1. Answering RQ1 & RQ2: The Architectural Shift from Cascaded Pipelines to VLMs

Regarding the first two research questions—concerning extraction accuracy and semantic comprehension—the results from Task 2 clearly indicate a paradigm shift in document understanding. The end-to-end VLMs outperformed the traditional two-step pipeline (LightOnOCR combined with LLMs) in extracting domain-specific medical entities and generating high-quality summaries.

This superiority can directly be attributed to the elimination of cascading errors. In the decoupled pipeline, any text missed, misread, or structurally garbled by the OCR engine forms an unrecoverable bottleneck; the subsequent LLM operates blindly on flawed textual input, severely limiting semantic comprehension. Conversely, VLMs inherently ground their semantic reasoning in the visual features of the document. By jointly attending to text, layout, and visual context in a single pass, models like Qwen3-VL and Pixtral-8B can correctly infer ambiguous medical terminology that a pure OCR engine would fail to digitize. The fact that Qwen3-VL achieved the highest overall Medical Term Recall (80.85%) explicitly answers the first research questions, underscoring the accuracy advantage of this multimodal approach for complex, unstandardized clinical records.

4.2. Answering RQ3: Reliability, Hallucinations, and Extraction Consistency

The third research question addressed how effectively the different approaches mitigate hallucinations and ensure reliability in critical domains. Despite the architectural advantages of VLMs in general accuracy, the evaluation exposed a critical vulnerability: extreme variance in extraction consistency. Across all evaluated VLMs, the standard deviation for Medical Term F1-scores consistently exceeded 30%, with individual sample performances oscillating between 0.0% and 100.0%.

This lack of determinism poses a significant challenge for autonomous robotic systems in healthcare. While a model might flawlessly extract all relevant data from a cleanly structured laboratory report, it may fail entirely or hallucinate values on a visually noisy patient discharge summary. This erratic behavior suggests that current VLMs, when deployed in a zero-shot setting, are highly sensitive to layout variations and prompt phrasing. To effectively mitigate these hallucinations (as posed by RQ3), automated extraction cannot yet operate fully unsupervised. A "Human-in-the-Loop" verification step—as successfully demonstrated in the interactive proof-of-concept—is strictly necessary to intercept catastrophic failures before the data enters a hospital's central system.

4.2.1. Deployment Constraints: Privacy over Peak Performance

A central contribution of this work is the translation of theoretical benchmark performance into a practical proof-of-concept for future robotic deployment. While Qwen3-VL achieved the highest theoretical extraction metrics, selecting a model for the target robotic platform required a compromise between peak algorithmic performance, hardware constraints, and legal compliance.

Clinical data is subject to the General Data Protection Regulation (GDPR), which imposes severe restrictions on data processing and sovereignty. Because the target mobile robot will lack the onboard computational capacity to run large multimodal models locally at sufficient speeds, cloud-based offloading is mandatory. In this context, utilizing Pixtral-8B via Mistral AI’s European-hosted enterprise API emerged as the most viable solution for the prototype. It offered performance closely trailing the benchmark leader while guaranteeing that sensitive patient data remains within European jurisdictions. This highlights that in clinical robotics, data privacy and system architecture dictate model selection just as heavily as raw F1-scores.

4.3. Limitations and Future Work

The findings of this study are subject to certain limitations that provide avenues for future research. First, the evaluation was conducted on a stratified subset of $N = 48$ documents in a strictly zero-shot setting. Providing the VLMs with few-shot examples—specifically tailored to the layout of standard hospital forms—could significantly reduce the observed variance and stabilize the extraction process.

Second, future iterations of the robotic platform could benefit from advancements in edge computing and model quantization. As smaller VLMs become more efficient, deploying models locally on the robot’s onboard hardware would entirely eliminate the latency and privacy concerns associated with cloud APIs. Finally, implementing a self-reflection loop, where the VLM assesses its own extraction confidence and actively triggers a fallback routine if it detects visual ambiguity, would greatly enhance the system’s autonomy and safety in real-world hospital environments.

5. Conclusions

This thesis presented the development, evaluation, and deployment of an automated information extraction system designed to enable a mobile robotic platform to digitize and structure complex clinical documents. By systematically comparing a traditional, decoupled OCR-LLM pipeline against state-of-the-art, end-to-end Vision-Language Models (VLMs), this work provided critical insights into the current capabilities and limitations of multimodal AI in healthcare environments.

The evaluation of Task 1 (raw text transcription) demonstrated that while standalone OCR engines like LightOnOCR provide reliable baselines, VLMs—particularly Pixtral-8B and Qwen3-VL—achieve superior character and word-level accuracy by contextually interpreting visually degraded or complex layouts. This advantage became even more pronounced in Task 2 (clinical information extraction). The end-to-end VLM architecture successfully bypassed the compounding cascading errors inherent to the two-step pipeline, resulting in noticeably higher Medical Term Recall and overall extraction F1-scores. Qwen3-VL emerged as the algorithmic frontrunner, achieving a Medical Term F1-score of 64.14%, closely followed by Pixtral-8B.

However, the study also revealed a significant operational hurdle: extreme performance variance. While VLMs can perfectly parse cleanly formatted records, their zero-shot extraction consistency drops drastically on noisy or unstandardized documents, leading to sample F1-scores oscillating between 0.0% and 100.0%. Although preliminary experiments showed that few-shot prompting can mitigate formatting failures and improve performance by 1-2%, the underlying sensitivity to layout shifts remains a risk for fully unsupervised robotic operation.

Ultimately, the successful design of this system required balancing peak algorithmic performance with strict practical constraints. Due to the target robot’s anticipated hardware limitations and the uncompromising data privacy regulations (GDPR) governing clinical records, offloading inference to a European-hosted enterprise API was identified as the necessary architectural path. Consequently, Pixtral-8B was selected as the final production model for the proof-of-concept demonstration. It offered the best compromise: providing near state-of-the-art multimodal extraction accuracy while ensuring absolute data sovereignty through Mistral AI’s European infrastructure.

In conclusion, end-to-end Vision-Language Models represent a significant leap forward for robotic document understanding. While further advancements in edge computing and model quantization are needed to bring these large models directly onto mobile hardware, the cloud-based Pixtral-8B integration developed and demonstrated in this thesis successfully validates the feasibility of autonomous, GDPR-compliant clinical data extraction for future healthcare robotics.

A. Appendix

A.1. System Prompts

To ensure full reproducibility, the following sections detail the exact German zero-shot system prompts utilized across the different experimental pipelines.

A.1.1. Task 1: VLM Pure OCR Extraction Prompt

This prompt was used in Task 1 to constrain Vision-Language Models to act strictly as optical character recognition systems, preventing generative hallucinations.

```
EXTRACTION_PROMPT = ""Du bist ein OCR-System fuer medizinische
Pflegedokumentation. Extrahiere den Text aus dem Bild so exakt
wie moeglich.
WICHTIGE REGELN:
- Erfinde keine neuen Informationen.
- Fasse nicht zusammen und interpretiere nicht.
- Uebernimm Zahlen, Masseinheiten, Uhrzeiten, Medikamentennamen
und Eigennamen exakt.
- Behalte Zeilenumbrueche und Reihenfolge so gut wie moeglich bei.
- Wenn ein Wort unklar ist, schreibe [UNK] statt zu raten.
Gib ausschliesslich den extrahierten Rohtext zurueck.""
```

A.1.2. Task 2: Approach 1 (OCR + LLM) Prompt

This prompt was used to guide the pure Large Language Models (Qwen3-8B, Ministral-8B, GPT-4o-mini) to summarize and extract entities from the noisy text strings generated by the external OCR engines.

```
"system": ""Du bist ein medizinischer Assistent, spezialisiert auf
Pflegedokumentation und die Fehlerkorrektur von fehlerhaften
OCR-Texten. Deine Aufgabe ist es, unsaubere Pflegenotizen zu lesen,
OCR-Tippfehler implizit zu korrigieren und die klinischen Fakten
strikt beizubehalten, ohne neue Informationen zu erfinden.""
```

```
"task": ""Analysiere den folgenden Text, der aus einer fehlerhaften
OCR-Texterkennung stammt. Fasse den Text zusammen und extrahiere die
wichtigsten Merkmale.
```

```
WICHTIGE REGELN:
- Bewahre ALLE medizinischen Fachbegriffe, Vitalwerte (z.B. Puls,
Atemfrequenz) und Pflegemaßnahmen (z.B. Wundversorgung, Dekubitus).
- Wenn ein Wort durch OCR falsch geschrieben ist (z.B. 'D3kubitus'),
korrigiere es stillschweigend zu 'Dekubitus'.
- Erfinde KEINE allgemeinen Begriffe wie 'Behandlung' oder
'Körperpflege', wenn sie nicht im Text stehen.
- Bleibe zu 100% bei den Fakten des Textes.
```

Strukturiere deine Antwort exakt wie folgt:

1. ****Zusammenfassung****: Kurze, sachliche Zusammenfassung der Notiz.
2. ****Wichtige Merkmale****: Stichpunktartige Liste der wichtigsten medizinischen Parameter und Symptome (exakte Fachbegriffe).

Text:

{text}"""

A.1.3. Task 2: Approach 2 & 3 (End-to-End VLM) Prompts

The following prompts illustrate the iterative refinement of the instructions provided to the Vision-Language Models. Version 1 was used initially, while Version 2 represents the final, optimized prompt used for the main evaluation.

Prompt Version 1 (Initial Iteration with Transcription Step)

"system": """Du bist ein KI-Experte für medizinische Pflegedokumentation. Deine Aufgabe ist es, handschriftliche Notizen direkt aus Bildern zu lesen, den medizinischen Kontext zu verstehen, OCR-Fehler kontextuell zu korrigieren und die klinischen Fakten strikt beizubehalten, ohne neue Informationen zu erfinden."""

"task": """Analysiere das beigefügte Bild der Pflegenotiz. Das Bild enthält handschriftlichen Text, der teilweise schwer lesbar sein kann.

WICHTIGE REGELN:

- Bewahre ALLE medizinischen Fachbegriffe, Vitalwerte (z.B. Puls, Atemfrequenz) und Pflegemaßnahmen (z.B. Wundversorgung, Dekubitus).
- Wenn ein Wort im Bild schwer lesbar ist, nutze den medizinischen Kontext zur logischen Korrektur.
- Erfinde KEINE allgemeinen Begriffe wie 'Behandlung' oder 'Körperpflege', wenn sie nicht im Bild stehen.
- Wenn ein Teil völlig unleserlich ist, rate nicht, sondern ignoriere ihn für die Merkmale.

Strukturiere deine Antwort exakt wie folgt:

1. ****Transkription****: [Schreibe hier den erkannten Text aus dem Bild auf. Korrigiere offensichtliche Fehler direkt stillschweigend.]
2. ****Zusammenfassung****: [Kurze, sachliche Zusammenfassung in 1-2 Sätzen.]
3. ****Wichtige Merkmale****:
 - [Parameter/Symptom]: [Exakter Wert oder Beschreibung aus dem Bild]"""

Prompt Version 2 (Final Optimized End-to-End Iteration)

"system": """Du bist ein erfahrener KI-Assistent für medizinische Pflegedokumentation. Du liest Pflegenotizen direkt aus Bildern und erstellst strukturierte klinische Zusammenfassungen.

GRUNDREGELN:

- Antworte AUSSCHLIESSLICH auf Deutsch, unabhängig von der Bildqualität.

- Behalte ALLE medizinischen Werte exakt bei: Vitalwerte, Medikamente mit Dosierung, Uhrzeiten, Mengenangaben.
- Erfinde KEINE Informationen, die nicht im Bild stehen.
- Auch bei schlechter Bildqualität: Erstelle eine Zusammenfassung mit dem was lesbar ist.""",

"task": ""Analysiere das beigefügte Bild einer handgeschriebenen oder gedruckten Pflegenotiz.

Strukturiere deine Antwort EXAKT in diesem Format | ohne Abweichungen:

****Zusammenfassung**:** [3-5 vollständige Sätze. Beschreibe chronologisch: Allgemeinzustand des Patienten, durchgeführte Pflegemaßnahmen, besondere Ereignisse oder Verhaltensauffälligkeiten, relevante Beobachtungen und Planungen. Nenne den Patienten beim Namen wenn lesbar.]

****Wichtige Merkmale**:**

- [Parameter/Symptom]: [Exakter Wert oder Beschreibung aus dem Bild | z.B. "RR 120/80 mmHg", "Puls 80/min", "Medikament X 5mg um 8 Uhr", "Trinkmenge 500ml", etc. | nur wenn erwähnt]

WICHTIG: Lasse Merkmale komplett weg wenn sie in der Notiz nicht vorkommen. Füge keine Merkmale hinzu die nicht im Bild stehen.""

A.2. Evaluation Dataset (N=48)

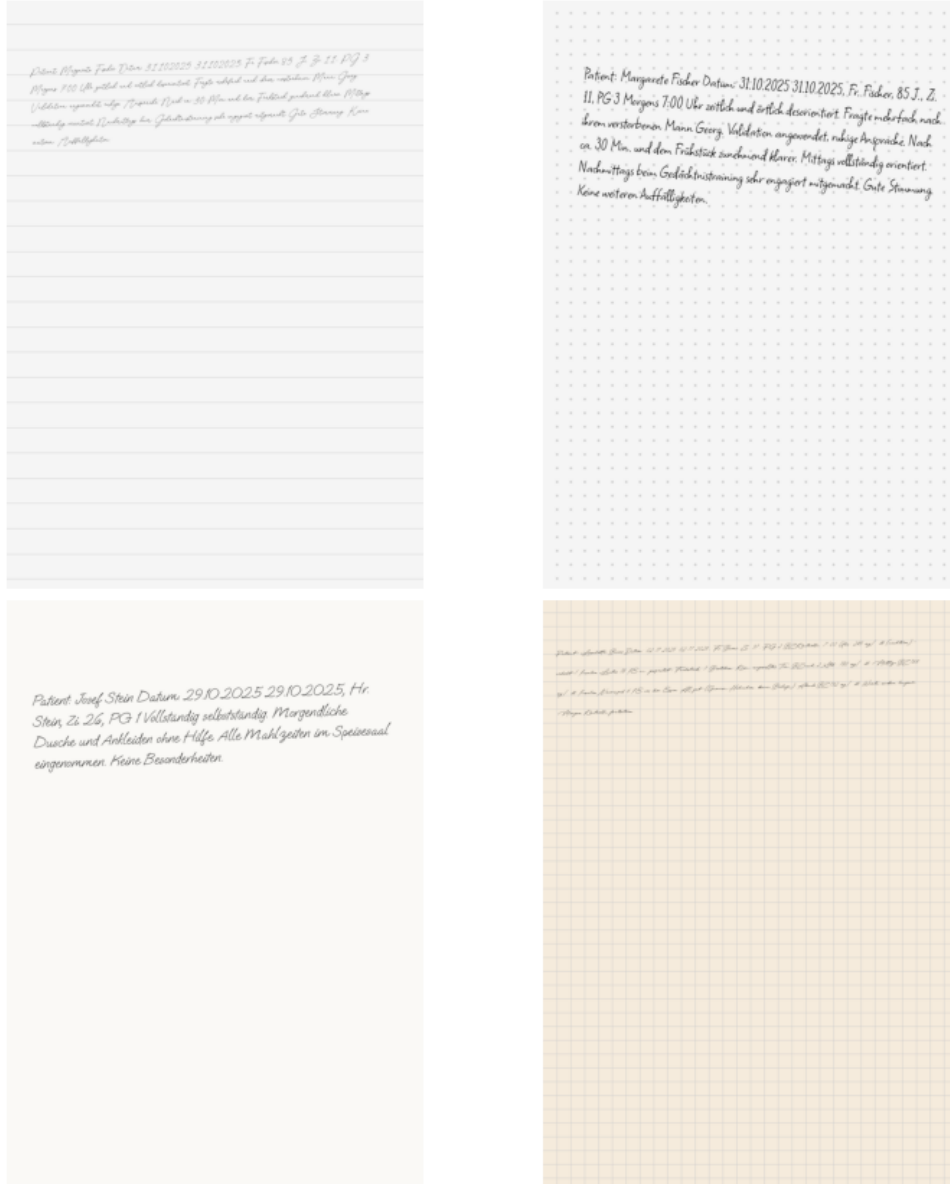


Figure A.1.: Samples 1–4 of the $N = 48$ evaluation subset, demonstrating full font coverage and scenario distribution.



Figure A.2.: Samples 5–8 of the $N = 48$ evaluation subset, demonstrating full font coverage and scenario distribution.



Figure A.3.: Samples 9–12 of the $N = 48$ evaluation subset, demonstrating full font coverage and scenario distribution.



Figure A.4.: Samples 13–16 of the $N = 48$ evaluation subset, demonstrating full font coverage and scenario distribution.



Figure A.5.: Samples 17–20 of the $N = 48$ evaluation subset, demonstrating full font coverage and scenario distribution.

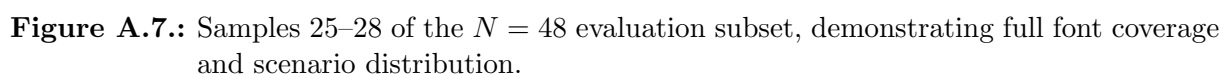


Figure A.8.: Samples 29–32 of the $N = 48$ evaluation subset, demonstrating full font coverage and scenario distribution.

Patient: Heinrich Bauer Datum
29.10.2025 29.10.2025, Hr. Bauer, 84 J., Ji.
50, PH 5 NACH MANDESEE!
Nachtdienst berichtete: - 23.30 Uhr im
Flur gefürden, suchte Toilette - 01.45
Uhr erneut unterwegs, wollte zur
Arbeit - 03.30 Uhr wieder wach, sehr
desorientiert Jagdienst. Hr. Bauer
müde und erschöpft, schlief bis 10.00
Uhr. Maßnahmen besprochen: -
Toilettengang vor dem Schlafen -
Nachtleicht im Zimmer -
Orientierungshilfen (Große Uhr,
Kalender) - Ggf. Medikation
überprüfen Dr. Hartmann für
Besprechung informiert.

70

[illegible]

Patient: Bruchhilde Wölf Datum: 30.03.2025, 30.03.2025, Fr
Wölf, 81 Jg, ZN 33, Pk 3 Seit 2 Tagen HbE/STN mit
gelblichem Auswurf. Heute Morgen klappte sie über
Atemnot beim Essen. Vitalenkontrollen 1000 l/min - Temp.
37,4°C - SpO2 94% (Raumluft) - FIO 22/min - beschleunigte
- RR 135/85%, Puls 96/min Auskultation: Rasselgeräusche
rechts basale. Dr. Lehmann 10:00 Uhr informiert, kam zur
Visite 10:30 Uhr. Diagnose: Bronchien, V.a. Pneumonie.
Erregern-Thorax angeordnet. Therapie: Amoxicillin 1000mg
5x tgl, ACC 600mg 5x tgl, Inhalation mit Salbutamol 1.
Stufe 1000 l/min Inhalation durchgeführt. Pat. kesselt
produktiv ab. Wissen: SpO2-Überwachung, Atemfrequenz
dokumentieren, Oberkörper hochlagern.

Figure A.11.: Samples 41–44 of the $N = 48$ evaluation subset, demonstrating full font coverage and scenario distribution.

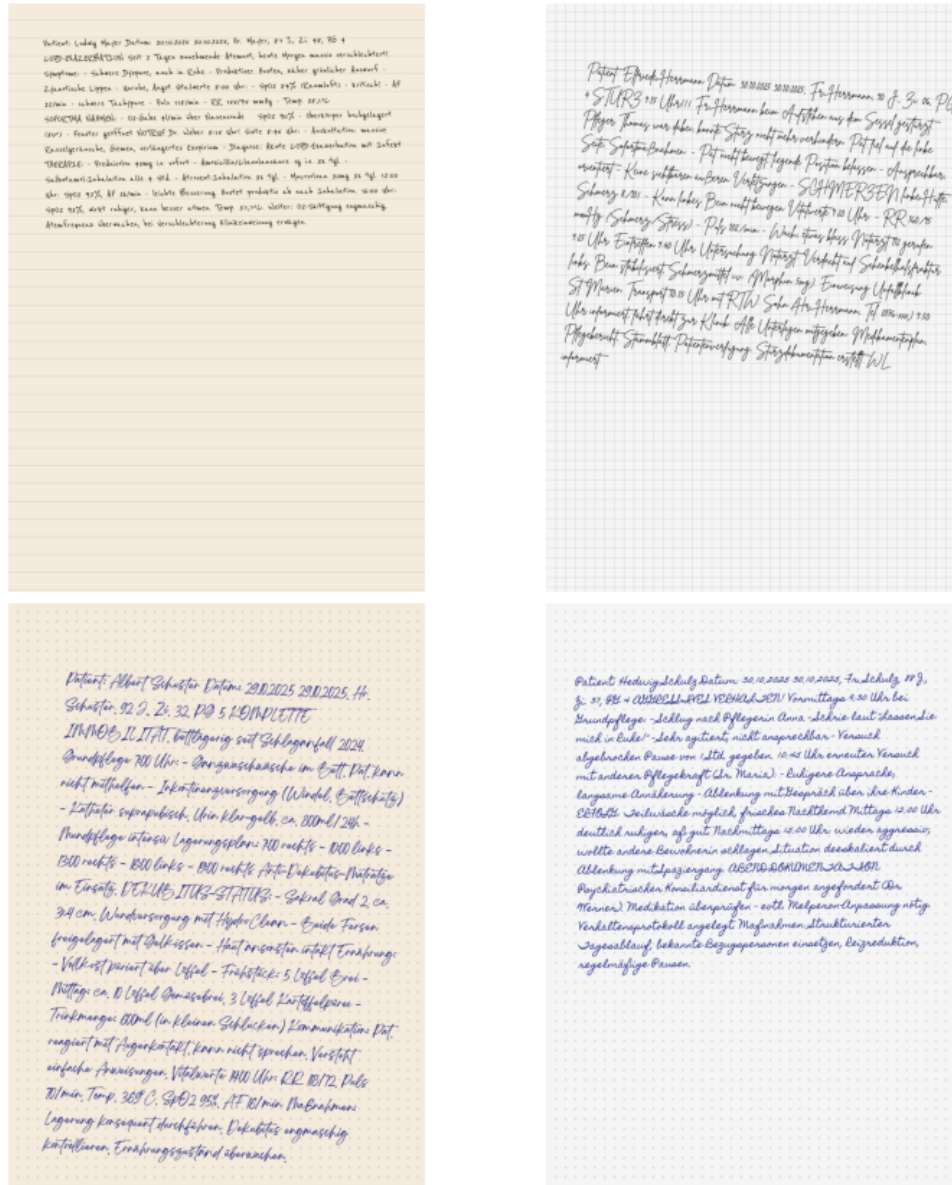


Figure A.12.: Samples 45–48 of the $N = 48$ evaluation subset, demonstrating full font coverage and scenario distribution.

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